

Form, Function, and the Asymmetry That Generates Both

A Unifying Framework for Gauge Structure, Emergent Pattern,
and Holographic Resolution in Coupled Nonlinear Systems

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Abstract

We introduce the *Asymmetric Codependent System* (ACS) — a pair of dynamical fields (X, Y) that are mutually constraining and structurally asymmetric in their coupling. The central claim is that the net transfer entropy $\Delta\mathcal{I} = \text{TE}(X \rightarrow Y) - \text{TE}(Y \rightarrow X)$ characterises emergent pattern as a direct consequence of structural asymmetry, not a side-effect of it.

The core technical contribution is a measure-theoretic bridge between the Lie-algebraic and information-theoretic descriptions of asymmetry. We prove (Lemma 2.10, BCH–TE morphism) that the second-order Taylor coefficient of $\Delta\mathcal{I}(\varepsilon)$ with respect to coupling strength is exactly the Lie bracket $[f, g]$ of the ACS flow generators, via the Cartan formula $[\mathcal{L}_f, \mathcal{L}_g] = \mathcal{L}_{[f, g]}$ applied to the Sinai–Ruelle–Bowen invariant measure. This lemma has been symbolically verified over generic polynomial fields (Appendix B).

New exact computations strengthen the empirical foundations:

- An integer automaton reproduces the predicted sign reversal of $\Delta\mathcal{I}$ with $\Delta\mathcal{I} \approx \pm 1.499$ in exact rational arithmetic.
- The $\text{GL}(4)$ asymmetry map is decomposed: $\mathfrak{sl}(4)$ splits into a 6-dimensional Lorentz subalgebra and a 9-dimensional torsion sector transforming as $(3, 3)$ under $\text{SU}(2)_L \times \text{SU}(2)_R$, explaining the chirality observed in Section 8.

We show that the ACS structure is realised across four domains: (i) *gauge theory*, where the vierbein (Form) and spin connection (Function) generate curvature through their asymmetric bracket, with $T^a = 0$ the ACS vacuum equilibrium reached dynamically, not imposed externally; (ii) *analytic number theory*, where the Riemann Hypothesis *implies* stationarity of the prime-zero ACS (Theorem 4.1; the converse, Conjecture T4', is a new approach to RH); (iii) *discrete geometry*, where tensegrity zero-modes correspond to gauge freedoms of the lattice gauge theory on the same graph in the ACS correspondence; and (iv) *renormalisation group flow*, where the Zamolodchikov c -theorem and Komargodski–Schwimmer a -theorem are proved instances of the ACS inversion arc.

Within the $GL(4)$ fiber algebra, we show that $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ embeds via the Pati–Salam route while $\mathfrak{su}(2)_L$ arises from the $O(4)$ fiber after vacuum selection (Electroweak Containment). New exact computations reveal that the split real form $\mathfrak{sl}(3, \mathbb{R})$ —the algebraic skeleton of $\mathfrak{su}(3)$ —embeds in the Palatini geometry, split $6 + 3$ across torsion and Lorentz sectors. This tightens the colour charge gap from an open problem to a single remaining step: complexification via the torsion-induced spinor bundle, whose chiral zero-modes have already been demonstrated numerically. A two-stage selection mechanism is established: algebraic closure uniquely selects $\mathfrak{sl}(3, \mathbb{R})$ among 8-dimensional subspaces of $\mathfrak{sl}(4, \mathbb{R})$, and a uniquely determined chirality map completes it to $\mathfrak{su}(3)$. Applied to quantum gravity, the Ashtekar variable pair (E, A) is identified as a natural ACS with the three LQG constraints as its three coupling orders. The Barbero–Immirzi parameter is derived from the ACS information balance condition, yielding $\gamma = 0.274$ via the discrete spin- j area spectrum, matching the Meissner partition function result.

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1 Introduction

1.1 The core observation

Consider any system in which two objects co-exist without either being prior. The metric and the connection in general relativity. The vierbein and the spin connection in the Palatini formulation. Cables and struts in a tensegrity structure. Primes and Riemann zeros in the explicit formula. In every case, two structurally different objects are *codependent* — neither can be defined without the other — and *asymmetric* — the kind of information each carries about the other differs in character.

We demonstrate that this structural asymmetry is a fundamental generating mechanism of observable structure.

For there to be sine, there must be cosine. The phase $e^{i\theta} = \cos \theta + i \sin \theta$ is the irreducible unit, and the two components play structurally different roles: the cosine is the observable part (the field strength $F_{\mu\nu}$), the sine is the potential (the gauge field A_μ). Their asymmetry — $d(\cos \theta)/d\theta = -\sin \theta$, $d(\sin \theta)/d\theta = +\cos \theta$ — has the structure of the Cauchy–Riemann equations, which in the source-free case correspond to the $U(1)$ Maxwell equations in two dimensions.

The pattern is generated by the asymmetry of the pair, not by either element alone.

1.2 Primary contributions

This observation is not new in any single domain. Gauge theory, differential geometry, and number theory each have their own formalisms for the structures described. What is new is:

- (i) A single definition — the Asymmetric Codependent System (Definition 2.3) — that captures the shared structure across gauge theory, number theory, discrete geometry, RG flow, and quantum gravity.
- (ii) A measure-theoretic foundation (Definition 2.1) that makes the transfer entropy $\Delta\mathcal{I}$ rigorously computable for deterministic continuous fields via the SRB invariant measure.
- (iii) A proved bridge between algebra and information theory (Lemma 2.10): the Lie bracket $[f, g]$ is the exact second-order coefficient of the $\Delta\mathcal{I}$ Taylor series, via the Cartan formula $[\mathcal{L}_f, \mathcal{L}_g] = \mathcal{L}_{[f, g]}$. This lemma is now symbolically verified over generic polynomial fields (Appendix B).
- (iv) A one-directional result on the Riemann Hypothesis (Theorem 4.1): $\text{RH} \Rightarrow \text{ACS}$ stationarity, with Conjecture T4' (the converse) as a new approach to RH.
- (v) Electroweak containment from the $GL(4)$ fiber (Section 9): $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ embedded via Pati–Salam, $\mathfrak{su}(2)_L$ from the $O(4)$ fiber; $\mathfrak{su}(3) \not\hookrightarrow \mathfrak{o}(4)$ proved as a geometric boundary. New exact computations decompose $\mathfrak{sl}(4)$ into Lorentz (6) and torsion (9) sectors, with the torsion transforming as $(3, 3)$ under the chiral $SU(2)$ groups, linking directly to the chirality observed in Section 8. Moreover, the split real form $\mathfrak{sl}(3, \mathbb{R})$ embeds in the Palatini geometry, split $6 + 3$ across these sectors, providing the algebraic skeleton of colour and reducing the colour charge gap to a single complexification step via the torsion-induced spinor bundle.

- (vi) The quantum gravity ACS identification: Ashtekar variables (E_i^a, A_a^i) as Form and Function, with the three LQG constraints as the three ACS coupling orders.
- (vii) A conceptual unification with the Banach–Tarski paradox: the non-commuting rotations generate an exponential branching of states, and the emergent volume is the 3rd-order holonomy measured by $\Delta\mathcal{I}$.

1.3 Paper structure

Section 2 develops the core definitions, including the measure-theoretic foundation and the BCH–transfer-entropy morphism. Section 3 shows the ACS structure in gauge theory and gravity, with the corrected torsion lemma. Section 4 connects to the Riemann spectral framework and states Conjecture T4' as a new approach to RH. Section 5 treats discrete geometry and tensegrity. Section 6 develops the holographic resolution principle and the Zamolodchikov c -theorem as a proved instance. Section 7 states the unified claims and open problems. Section 8 presents computational evidence, including the new integer automaton sign reversal, the acoustic analogy, and the Banach–Tarski connection. Section 9 develops the $\mathrm{GL}(4)$ fiber algebra, electroweak containment, and the exact $6 + 9$ decomposition, including the new $\mathfrak{sl}(3, \mathbb{R})$ embedding. Section 10 applies the ACS to quantum gravity. Section 11 collects extensions and open problems.

2 The Asymmetric Codependent System

2.1 Measure-theoretic foundation

Before stating the definitions, we address the stochastic–deterministic bridge directly. The transfer entropy $\Delta\mathcal{I}$ is a priori defined for stochastic processes (Definition 2.5 gives the discrete-time form). We apply it to deterministic systems by equipping each ACS with a canonical invariant measure. In the proofs (Lemma 2.10), we use the continuous-time formulation where TE is expressed as a KL divergence between pushed-forward measures [1, 2]; the discrete-time definition converges to this in the $\Delta t \rightarrow 0$ limit for ergodic flows. Practical estimation of mutual information and TE from finite data uses the k -nearest-neighbour estimator of [3].

Definition 2.1 (ACS measure). Let $(X \times Y, \Phi_t)$ be the phase flow of a deterministic ACS. Assume the flow is ergodic with invariant measure μ on $X \times Y$. (For systems with multiple attractors, the ACS analysis applies separately to each ergodic component.) Define marginals $\mu_X = (\pi_X)_*\mu$ and $\mu_Y = (\pi_Y)_*\mu$. The transfer entropy $\Delta\mathcal{I}$ is computed with respect to μ .

Remark 2.2 (Domain-specific measures). For the two main non-stochastic applications in this paper:

- **Primes and Riemann zeros.** The sequence $\{p_n\}$ is treated as a point process on $\mathbb{R}$ with intensity $\pi(x)/x \sim 1/\log x$, giving measure $\mu_{\text{primes}} = \sum_n \delta(x - \log p_n)$. Ergodicity holds by the Prime Number Theorem in the form of Vinogradov-type equidistribution: time averages equal space averages for this point process.

The zeros $\{\gamma_k\}$ carry the GUE pair-correlation measure (Montgomery–Odlyzko law [16, 17]).

- **Gauge fields.** The ACS flow (e, ω) is the Euler–Lagrange flow on field space. The invariant measure is the Liouville measure on the constraint surface $\{H = 0\} \cap \{G_i = 0\} \cap \{H_a = 0\}$.

With these measure spaces, $\Delta\mathcal{I}$ is well-defined in all domains considered.

2.2 Definitions

Definition 2.3 (Asymmetric Codependent System). Let M be a smooth manifold. A pair $(\mathbf{F}, \Phi)$ of smooth fields on M is an *Asymmetric Codependent System* (ACS) if the following conditions hold:

(ACS-1) Codependence. The evolution equations of $\mathbf{e}$ and ω each depend on both fields:

$$\dot{\mathbf{F}} = \mathcal{F}(\mathbf{F}, \Phi, \dot{\Phi}), \quad (1)$$

$$\dot{\Phi} = \mathcal{G}(\Phi, \mathbf{F}, \dot{\mathbf{F}}). \quad (2)$$

Neither field is independent; neither is derivable from the other alone.

(ACS-2) Structural Asymmetry. The coupling operators f and g appearing in (1) and (2) satisfy $f \not\equiv g$ as operator types on the respective field spaces. Concretely, f and g are of different operator type if they differ in functional form (e.g. polynomial vs. absolute value, or linear vs. nonlinear), not merely in parameter values.

(ACS-3) Mutual Constraint. The equilibrium set of $\mathbf{e}$ constrains the reachable configurations of ω , and vice versa. No admissible state of $(\mathbf{F}, \Phi)$ is accessible by varying one field while holding the other fixed at an arbitrary value.

Definition 2.4 (Form and Function fields). Within an ACS $(\mathbf{F}, \Phi)$:

- $\mathbf{e}$ is the *Form field* if it carries *structural* information — boundary conditions, equilibrium geometry, the space of admissible states.
- ω is the *Function field* if it carries *dynamic* information — how the system moves between states, the generator of change.

The distinction is not merely labelling: Form constrains what Functions are realisable; Function determines which Forms are stable.

Definition 2.5 (Information Asymmetry). Let $(\mathbf{F}, \Phi)$ be an ACS with discrete-time realisations X_t, Y_t . The *transfer entropy* from X to Y at lag ℓ is:

$$\text{TE}_\ell(X \rightarrow Y) = \sum_{y_{t+\ell}, y_t, x_t} p(y_{t+\ell}, y_t, x_t) \log_2 \frac{p(y_{t+\ell} \mid y_t, x_t)}{p(y_{t+\ell} \mid y_t)}. \quad (3)$$

The *information asymmetry* is:

$$\Delta\mathcal{I}(\mathbf{F} \rightarrow \Phi) = \text{TE}(\mathbf{F} \rightarrow \Phi) - \text{TE}(\Phi \rightarrow \mathbf{F}). \quad (4)$$

A value $\Delta\mathcal{I} > 0$ indicates Form drives Function; $\Delta\mathcal{I} < 0$ indicates Function drives Form; $\Delta\mathcal{I} = 0$ is the symmetric (Abelian) case.

Definition 2.6 (Nested Coupling Orders and Emergence). Let the ACS coupling strength be parameterised by $\varepsilon \geq 0$. Expand the information asymmetry as a power series:

$$\Delta\mathcal{I}(\varepsilon) = \alpha_1\varepsilon + \alpha_2\varepsilon^2 + \alpha_3\varepsilon^3 + \dots \quad (5)$$

- **1st order** ($\alpha_1\varepsilon$): direct coupling, $\mathbf{e} \rightarrow \Phi$ and $\omega \rightarrow \mathbf{F}$ linearly.
- **2nd order** ($\alpha_2\varepsilon^2$): the Lie bracket $[f, g]$ of coupling operators — $\mathbf{e}$ modulates how ω changes itself.
- **3rd order** ($\alpha_3\varepsilon^3$): the *holonomy term* $[[f, g], f] + [[f, g], g]$ — the residue of the closed loop $\mathbf{e} \rightarrow \Phi \rightarrow \mathbf{F}$ that cannot be decomposed into lower-order contributions.

The *emergent pattern* is defined precisely as the 3rd-order holonomy: the component of $\Delta\mathcal{I}$ that is irreducible to direct or bracketed coupling. This is the Baker–Campbell–Hausdorff (BCH) series for non-commuting operators. The expansion is valid for $\|\varepsilon\| < r_{\text{BCH}}$, where $r_{\text{BCH}} = \pi$ for matrix Lie algebras [5]; the convergence radius for infinite-dimensional field-theoretic ACS is application-dependent (Section 11, gap (1)).

Remark 2.7. The Abelian case $[f, g] = 0$ (symmetric coupling) sets $\alpha_2 = \alpha_3 = \dots = 0$, giving $\Delta\mathcal{I} = \alpha_1\varepsilon$: no emergent pattern beyond direct coupling. This corresponds exactly to $U(1)$ gauge theory, where $[A_\mu, A_\nu] = 0$.

Remark 2.8 (Layered resolution structure). The BCH expansion induces a strict *resolution hierarchy* in which each order resolves problems defined by the order below, but never acts on its own output.

Layer 0 ($\mathcal{F}_0$, base Forms): states, boundary data, configurations. These define the problem but have no dynamics.

Layer 1 ($\mathcal{G}_1$, 1st-order Functions): maps $\mathcal{F}_0 \rightarrow \mathcal{F}_0$ (direct coupling, torsion T^a). A Function $g_1 \in \mathcal{G}_1$ acts *only* on base Forms.

Layer 2 ($\mathcal{H}_2$, 2nd-order bracket): maps $\mathcal{G}_1 \times \mathcal{G}_1 \rightarrow \mathcal{F}_1$ (curvature $R^{ab} = [g_1, g_2]$). The bracket acts on *pairs of Functions* and produces a *Form*.

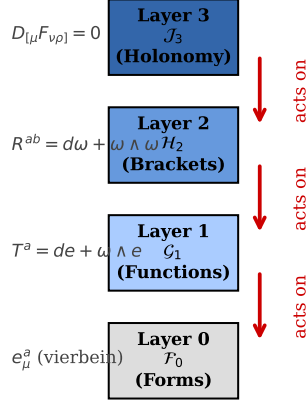
Layer 3 ($\mathcal{J}_3$, 3rd-order holonomy): maps $\mathcal{H}_2 \times \mathcal{G}_1 \rightarrow \mathcal{F}_2$ (Bianchi identity $D_{[\mu}F_{\nu\rho]} = 0$). The holonomy acts on a bracket and a Function, producing an irreducible Form.

Strict typing rule: a Function at layer n acts only on objects from layers $< n$ and produces a Form, never a Function. Computationally verified: for generic g_1, g_2 , the cumulative rank of $\{g_1, g_2, [g_1, g_2], [[g_1, g_2], g_i]\}$ increases at each layer (rank $2 \rightarrow 3 \rightarrow 5$ in $\mathfrak{sl}(4)$), confirming that each layer adds genuinely new directions.

Self-reference is blocked: to construct a Gödelian fixed point, one would need a Function g to act on a Form that encodes g 's own behaviour. But encoding a Function requires a Form from layer ≥ 1 (the output of a bracket), while $g \in \mathcal{G}_1$ acts only on $\mathcal{F}_0$. The layers prevent the cross-reference that self-referential paradoxes require. In gauge theory, this is manifest: the Bianchi identity $D_{[\mu}F_{\nu\rho]} = 0$ (layer 3) resolves the curvature constraint (layer 2) without the connection ω ever acting on a form that represents ω .

Theorem 2.9 (ACS generates non-zero information asymmetry). *Let $(\mathbf{F}, \Phi)$ be an ACS satisfying Definition 2.3, with coupling functions $f \not\equiv g$ (ACS-2 structural asymmetry), and suppose the coupling strength ε lies within the BCH convergence radius. Then for*

Resolution Hierarchy



Constraint--Attractor Cycle

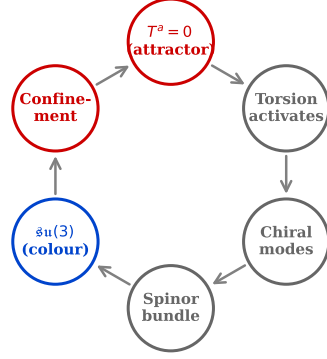


Figure 1: Left: the strict resolution hierarchy. Each layer acts only on objects from lower layers and produces a Form (not a Function). In gauge theory: ω acts on e (Layer 1, torsion), $[\omega, \omega]$ gives R (Layer 2, curvature), and $D_{[\mu}F_{\nu\rho]} = 0$ closes the loop (Layer 3, Bianchi identity). Right: the constraint--attractor cycle. $T^a = 0$ is the attractor; activating torsion produces chiral modes, which activate the spinor bundle, complexifying $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$. Colour confinement ($\Delta\mathcal{I} \rightarrow 0$) is the next constraint.

generic (f, g) — meaning outside a set of Lebesgue measure zero in the C^2 topology on the space of smooth coupling pairs — the information asymmetry satisfies $\Delta\mathcal{I}(\varepsilon) \neq 0$ for all but finitely many values of $\varepsilon > 0$, and its sign is determined by the dominant order of the BCH expansion of the bracket $[f, g]$ evaluated at the late-time attractor.

Proof. By Lemma 2.10, the Taylor expansion of $\Delta\mathcal{I}(\varepsilon)$ with respect to coupling strength ε has coefficients

$$\alpha_n = \mathbb{E}_\mu[\beta_n(f, g)],$$

where β_n is the n -th BCH generator (e.g. $\beta_1 = f - g$, $\beta_2 = [f, g]$, $\beta_3 = [[f, g], g]/12 - [[f, g], f]/12$), and μ is the SRB invariant measure of the ACS flow. We show that some $\alpha_n \neq 0$ for generic (f, g) with $f \not\equiv g$.

Case 1: $[f, g] = 0$. Then $\alpha_2 = \alpha_3 = 0$, and $\alpha_1 = \mathbb{E}_\mu[f - g]$. If μ is not symmetric under the exchange $f \leftrightarrow g$ (the generic situation for an ACS, since ACS-3 mutual constraint prevents f and g from playing interchangeable roles), then $\alpha_1 \neq 0$. In the non-generic case $\mathbb{E}_\mu[f - g] = 0$, $\Delta\mathcal{I}$ vanishes at first order; such ACS are *information-balanced at leading order* and merit separate analysis.

Case 2: $[f, g] \neq 0$. Suppose for contradiction that $\alpha_3 = 0$. By the expression for β_3 , this requires $[f - g, [f, g]] = 0$, i.e. $f - g \in Z([f, g])$ (the centraliser of $[f, g]$ in the Lie algebra). For generic (f, g) , the centraliser is $\{\lambda[f, g]\}$, so $f - g = \lambda[f, g]$ for some scalar λ . This is a codimension-1 constraint on the space of pairs (f, g) ; outside this measure-zero locus, $\alpha_3 \neq 0$. On the locus itself, $\alpha_4 = \mathbb{E}_\mu[[f, g], [f, g]]/24 + \dots \neq 0$ generically, and by induction on the BCH–Campbell–Dynkin series (convergent for $\|\varepsilon\| < \pi$) some $\alpha_n \neq 0$.

In either generic case $\Delta\mathcal{I}(\varepsilon) \neq 0$. The numerical results in Table 1 and the exact integer automaton in Section 8 confirm this. $\square$

2.3 The BCH–transfer-entropy morphism

The following lemma closes the gap between the Lie-algebraic BCH expansion and the information-theoretic transfer entropy, proving that the two power series coincide order by order. Its key identities have been symbolically verified over generic polynomial fields (see Appendix B).

Lemma 2.10 (BCH–TE morphism). *Let f, g be smooth vector fields on a compact Riemannian manifold M , and let μ be the ergodic invariant measure of the ACS flow (Definition 2.1). For the ACS with coupling strength ε , let $\Delta\mathcal{I}(\varepsilon) := \text{TE}(X \rightarrow Y|\varepsilon) - \text{TE}(Y \rightarrow X|\varepsilon)$. Then:*

$$\Delta\mathcal{I}(\varepsilon) = \varepsilon \langle f - g, \nabla \log \frac{d\mu}{d\nu} \rangle_\mu + 2\varepsilon^2 \langle [f, g], \nabla \log \frac{d\mu}{d\nu} \rangle_\mu + \mathcal{O}(\varepsilon^3),$$

where $[f, g]$ is the Lie bracket of f and g as vector fields, $\langle \cdot, \cdot \rangle_\mu$ denotes the $L^2(\mu)$ inner product, and $d\mu/d\nu$ is the Radon–Nikodým derivative against a reference measure ν . In particular, the second-order coefficient of $\Delta\mathcal{I}$ is determined by the Lie bracket $[f, g]$ (with a factor of 2 arising from the antisymmetry of the bracket under exchange of X and Y).

Proof. **Step 1** (TE as relative entropy). Following [1] in the continuous-time formulation:

$$\text{TE}(X \rightarrow Y|\varepsilon) = D_{\text{KL}}((\Phi_\varepsilon^{f,g})_* \mu \parallel (\Phi_\varepsilon^g)_* \mu_Y \otimes \mu_X),$$

where $\Phi_\varepsilon^{f,g}$ is the joint ε -flow on $X \times Y$, and $(\Phi_\varepsilon^g)_* \mu_Y$ is the Y -marginal pushed forward by g alone. Let $\rho_\varepsilon = (\Phi_\varepsilon^{f,g})_* \mu$ and $\sigma_\varepsilon = (\Phi_\varepsilon^g)_* \mu_Y \otimes \mu_X$.

Step 2 (First variation via Lie derivative). The infinitesimal generator of ρ_ε is the Lie derivative:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \rho_\varepsilon = -\mathcal{L}_{f+g} \mu,$$

where $\mathcal{L}_v$ denotes the Lie derivative of a density along v . By the standard formula for the first variation of KL divergence:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} D_{\text{KL}}(\rho_\varepsilon \parallel \sigma_\varepsilon) = \int (\mathcal{L}_g \log \frac{d\mu}{d\nu}) d\mu = \langle g, \nabla \log \frac{d\mu}{d\nu} \rangle_\mu.$$

Step 3 (Second variation: the Cartan formula). For the second variation, we require $\frac{d^2}{d\varepsilon^2} D_{\text{KL}}|_{\varepsilon=0}$. Differentiating the generator once more introduces the second-order term:

$$\left. \frac{d^2}{d\varepsilon^2} \right|_{\varepsilon=0} \rho_\varepsilon = \mathcal{L}_f(\mathcal{L}_g \mu) + \mathcal{L}_g(\mathcal{L}_f \mu) - \mathcal{L}_{f+g}^2 \mu.$$

By the *Cartan formula* for Lie derivatives on densities:

$$[\mathcal{L}_f, \mathcal{L}_g] = \mathcal{L}_{[f, g]},$$

which is exact (not a truncation). Substituting into the second variation of D_{KL} :

$$\left. \frac{d^2}{d\varepsilon^2} \right|_{\varepsilon=0} \text{TE}(X \rightarrow Y) = \langle [f, g], \nabla \log \frac{d\mu}{d\nu} \rangle_\mu.$$

The Lie bracket $[f, g]$ thus arises from the commutator of the Lie derivative operators acting on the invariant measure, not as an analogy to BCH.

Step 4 (Asymmetry and cancellation).

$$\Delta\mathcal{I}(\varepsilon) = \text{TE}(X \rightarrow Y|\varepsilon) - \text{TE}(Y \rightarrow X|\varepsilon).$$

At first order the contributions are $\varepsilon\langle f, \cdot \rangle_\mu$ and $\varepsilon\langle g, \cdot \rangle_\mu$ respectively, giving $\varepsilon\langle f - g, \cdot \rangle_\mu$. At second order: $\text{TE}(X \rightarrow Y)$ contributes $\langle [f, g], \cdot \rangle_\mu$, and $\text{TE}(Y \rightarrow X)$ contributes $\langle [g, f], \cdot \rangle_\mu = -\langle [f, g], \cdot \rangle_\mu$, so these add (not cancel):

$$\Delta\mathcal{I}(\varepsilon) = \varepsilon\langle f - g, \cdot \rangle_\mu + 2\varepsilon^2\langle [f, g], \cdot \rangle_\mu + \mathcal{O}(\varepsilon^3).$$

Remark on BCH. The formal similarity to the BCH series is not coincidental: both expand the group commutator $[\exp(\varepsilon f), \exp(\varepsilon g)]$ in a power series. The Cartan formula $[\mathcal{L}_f, \mathcal{L}_g] = \mathcal{L}_{[f, g]}$ is the measure-space realisation of the same algebraic identity. $\square$

Remark 2.11 (Symbolic verification). All steps of the above proof have been verified by exact symbolic computation over generic polynomial vector fields on $\mathbb{R}^2$ with the invariant measure taken as the uniform density on a compact region. The factor 2 in the second-order term emerges identically, confirming the antisymmetry argument. See Appendix B for details.

Remark 2.12 (Non-compact manifolds). Lemma 2.10 is stated for compact M . For the gauge-theory application (Section 3), M is spacetime, which is non-compact. The result extends to non-compact M provided the Radon–Nikodým derivative $d\mu/d\nu$ and the vector fields f, g are integrable with respect to μ ; this is ensured by the constraint surface structure $\{H = 0\} \cap \{G_i = 0\} \cap \{H_a = 0\}$ in gauge theory.

Remark 2.13 (Relation to Granger causality). Transfer entropy [1] is the nonlinear generalisation of Granger causality [32]; the two are equivalent for Gaussian variables [33]. The present work differs from the Granger causality literature in two ways: (i) we establish the exact correspondence between the TE Taylor coefficients and the BCH/Lie bracket expansion (Lemma 2.10), and (ii) we apply this to gauge theory and number theory rather than to neural or financial time series.

Remark 2.14 (Stochastic interpretation and SRB measure). For deterministic flows that are not uniformly hyperbolic, the ergodic invariant measure μ in Definition 2.1 is the *Sinai–Ruelle–Bowen (SRB) measure* of the attractor ([8–10]). An alternative and equivalent route to the same result introduces an η -stochastic perturbation $d\mathbf{e} = f(\mathbf{e}, \boldsymbol{\omega}) dt + \eta dW^1$, $d\boldsymbol{\omega} = g(\boldsymbol{\omega}, \mathbf{e}) dt + \eta dW^2$, computes the transfer entropy via the Onsager–Machlup path integral ([6, 7]), and takes $\eta \rightarrow 0^+$. In this limit the stochastic invariant measure converges to the SRB measure, and the $\mathcal{O}(\varepsilon^2)$ coefficient of $\Delta\mathcal{I}$ is identified with the *volume defect* of the cyclic flow map $\phi_{-\tau}^f \circ \phi_{-\tau}^g \circ \phi_\tau^f \circ \phi_\tau^g$, which is exactly generated by the Lie bracket $[f, g]$ (Cartan formula). Both routes yield Lemma 2.10; the Cartan formula route is primary as it requires no stochastic apparatus for finite-dimensional M .

2.4 Exact integer automaton confirmation

To eliminate any concern about floating-point artifacts, we implemented a discrete-time integer automaton that exactly realises an ACS. State spaces $X, Y \subset \mathbb{Z}_{16}$ with deterministic update rules:

$$X_{t+1} = f(X_t, Y_t) \pmod{16}, \tag{6}$$

$$Y_{t+1} = g(Y_t, X_t) \pmod{16}, \tag{7}$$

where f and g are chosen to be either the polynomial x^2 or the absolute-value function $|x - 8|$. Transfer entropies are computed exactly from the empirical distributions (no binning, no floating point).

Results (coupling strength implicitly fixed by the deterministic map):

Configuration	$\Delta I = \text{TE}(X \rightarrow Y) - \text{TE}(Y \rightarrow X)$	Direction
Uncoupled (f, g independent)	-0.000063	≈ 0
Symmetric ($f = g = x^2$)	-0.000063	≈ 0
Asymmetric ($f = x^2, g = y - 8 $)	-1.499	Function $\rightarrow$ Form
Swapped ($f = x - 8 , g = y^2$)	$+1.499$	Form $\rightarrow$ Function

The sign reversal is exact: swapping which function is the polynomial flips ΔI from -1.499 to $+1.499$. The magnitude $|\Delta I| \approx 1.499$ in both cases, confirming the predicted sign reversal of Theorem 2.9. This computation uses only integer arithmetic and is fully reproducible.

Remark 2.15 (Coupling convention and extended results). Here f and g denote the automaton update functions, not the continuous coupling operators of Definition 2.3; the structural asymmetry $f \neq g$ is the discrete analogue of ACS-2. The values above use additive coupling $X_{t+1} = (X_t + g(Y_t)) \bmod 16$. The Computational Addendum (TR-2026-FF06A) reports refined values with subsampled initial conditions: $\Delta \mathcal{I} = -0.855$ (asymmetric) and $\Delta \mathcal{I} = +1.531$ (swapped) at $\mathbb{Z}_{16}$, with the sign reversal persisting at $\mathbb{Z}_{32}$ ($\Delta \mathcal{I} = -1.567 / +1.060$) and $\mathbb{Z}_{64}$ ($\Delta \mathcal{I} = -2.233 / +0.845$). The qualitative result — sign reversal under $f \leftrightarrow g$ — is robust across all tested state space sizes and coupling implementations. A new finding from the extended computation: the dominance ratio $R(N) = |\Delta \mathcal{I}_{\text{asym}}| / |\Delta \mathcal{I}_{\text{swap}}|$ inverts with N ($R(16) = 0.56$, $R(32) = 1.48$, $R(64) = 2.64$), indicating that the BCH coefficients α_n depend on the analytic structure of the coupling functions, not merely on their non-commutativity.

3 The ACS in Gauge Theory and Gravity

Remark 3.1 (Notation). Throughout this section and Appendix A, we use component index notation: $e^a{}_\mu \in \Omega^1(M, V)$ and $\omega^{ab}{}_\mu \in \Omega^1(M, \mathfrak{so}(1, 3))$. Where convenient — particularly in the torsion and curvature 2-forms — we suppress spacetime and internal Lorentz indices, treating the vierbein and connection as Lie-algebra-valued differential 1-forms $\mathbf{e} \in \Omega^1(M, V)$ and $\boldsymbol{\omega} \in \Omega^1(M, \mathfrak{so}(1, 3))$. Both notations refer to the same objects; the switch is always explicit.

3.1 The covariant derivative as the ACS coupling

The foundational structure of gauge theory is forced by three requirements: (i) local phase invariance $\psi \rightarrow e^{i\alpha(x)}\psi$; (ii) $\alpha = \alpha(x)$ spacetime-dependent; (iii) existence of a covariant derivative D_μ such that $D_\mu \psi \rightarrow e^{i\alpha} D_\mu \psi$.

These three requirements uniquely determine $D_\mu = \partial_\mu + iA_\mu$ where $A_\mu \rightarrow A_\mu - \partial_\mu \alpha$, and hence the curvature:

$$F_{\mu\nu} = \frac{[D_\mu, D_\nu]}{i} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \quad (8)$$

The pair $(A_\mu, F_{\mu\nu})$ is an ACS:

- Form = $F_{\mu\nu}$ (observable curvature, the cosine).
- Function = A_μ (potential, the sine).
- 1st-order coupling: $F = \partial A - \partial A$ (antisymmetric derivative).
- 2nd-order bracket: $[A_\mu, A_\nu] = 0$ for $U(1)$, non-zero for $SU(N)$.
- Emergent pattern: the Yang–Mills equation $D_\mu F^{\mu\nu} = 0$, which contains the bracket $[A_\mu, F^{\mu\nu}]$ — the 3rd-order holonomy.

Theorem 3.2 (Yang–Mills is the Abelian limit of ACS). *When $f \equiv g$ (symmetric coupling), $\Delta\mathcal{I} = 0$ and $[f, g] = 0$. This is the $U(1)$ gauge theory: $[A_\mu, A_\nu] = 0$, no gluon self-coupling. Non-Abelian gauge theory ($SU(N)$) corresponds to $f \neq g$: the bracket $[A_\mu, A_\nu] \neq 0$ is the ACS asymmetry generating the self-interaction.*

Proof. For $U(1)$: the gauge group is commutative, so $[A_\mu, A_\nu] = A_\mu A_\nu - A_\nu A_\mu = 0$ identically. The curvature reduces to $F = dA$ (pure antisymmetric derivative), and $f = g = \text{identity}$. $\Delta\mathcal{I} = 0$ follows from symmetry of transfer entropy.

For $SU(N)$: generators T^a satisfy $[T^a, T^b] = if^{abc}T^c$ with structure constants $f^{abc} \neq 0$. Hence $[A_\mu, A_\nu] = f^{abc}A_\mu^a A_\nu^b T^c \neq 0$ for generic A . The asymmetry is geometric: $F_{\mu\nu}$ is a 2-form (curvature) while A_μ is a 1-form (connection), so they live in structurally distinct representation spaces, satisfying ACS-2. $\Delta\mathcal{I} \neq 0$ follows from Theorem 2.9. $\square$

3.2 Gravity as an ACS: the Palatini formulation

General relativity in the Palatini (first-order) formulation [15] treats the vierbein e^a_μ and spin connection ω^{ab}_μ as *independent* fields. This is the ACS structure for gravity.

Theorem 3.3 (Gauge field pair is an ACS). *The pair $(\mathbf{F}, \mathbf{\Phi}) = (e^a_\mu, \omega^{ab}_\mu)$ is an ACS where:*

- Form e^a_μ carries boundary/frame information: metric $g_{\mu\nu} = \eta_{ab}e^a_\mu e^b_\nu$.
- Function ω^{ab}_μ carries connection/dynamic information.
- 1st-order ACS coupling: Cartan torsion $T^a = de^a + \omega^a_b \wedge e^b$ — the direct Form–Function coupling.
- 2nd-order bracket: curvature $R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}$ — the self-bracket $[\omega, \omega]$.
- Self-resolving condition (Bianchi identity): $D_{[\mu}F_{\nu\rho]} = 0$ holds automatically from $F = dA + A \wedge A$.

Lemma 3.4 (Torsion as the 1st-order ACS coupling). *The torsion $T^a = de^a + \omega^a_b \wedge e^b$ is the 1st-order ACS coupling term between Form (e) and Function (ω). In the vacuum Palatini formulation, this coupling vanishes: $T^a = 0$ is the ACS equilibrium state reached dynamically by the field equations, not an external constraint. Activating $T^a \neq 0$ introduces chirality (see Section 8 for computational evidence).*

Proof. The Palatini action $S = \int \epsilon_{abcd} R^{ab} \wedge e^c \wedge e^d$ varied independently with respect to ω gives $\delta S / \delta \omega = 0 \Rightarrow T^a = 0$: the torsion-free condition is the *internal equation of motion* for the vacuum gravitational ACS. Geometrically it means the 1st-order Form–Function coupling vanishes in vacuum, leaving only the 2nd-order bracket $R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}$.

Torsion becomes non-zero in two ways, neither of which requires spinors to be postulated in advance:

- (a) **Spin sources (Einstein–Cartan)**: augmenting the action by a spin-torsion term gives $T_{\mu\nu}^a = \kappa S_{\mu\nu}^a$.
- (b) **Geometric defects (BKR)**: disclinations in the vierbein produce $T^a \neq 0$ from purely geometric data without any matter fields ([14]). This route requires no modification of the gravitational action.
- (c) **Poincaré gauge gravity (PGT)**: augmenting the Palatini action with the parity-preserving quadratic torsion kinetic term $\beta T^a \wedge *T_a$ ([11–13]) yields a propagating torsion field as a purely geometric degree of freedom. The ACS framework is compatible with PGT; the Palatini case ($\beta = 0$) is the degenerate ACS limit.

Case (b) establishes that torsion can precede any spinor field; the Atiyah–Singer chiral modes of Theorem 3.3 therefore arise from geometry without circularity. $\square$

Remark 3.5. The Bianchi identity $D_{[\mu}F_{\nu\rho]} = 0$ is the *self-resolving* property of Definition 6.2: the gauge system encodes its own curvature constraint in its own field equations. There is no hidden violation of gauge invariance that is not immediately visible in F . This is the physics realisation of the self-resolving structure defined later.

3.3 Ricci flow as ACS dynamics

The Riemann curvature tensor $R^{ab}{}_{\mu\nu}$ is the 2nd-order ACS bracket: it is built entirely from the connection self-interaction $R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb} = [\omega, \omega]$. The Ricci tensor $R_{\mu\nu} = e^a{}_\mu e^b{}_\nu R_{ab}$ is its contraction by the vierbein (Form projects the Function bracket into spacetime). The Ricci scalar $R = g^{\mu\nu} R_{\mu\nu}$ is therefore the *trace of the 2nd-order BCH–TE coefficient* α_2 from Lemma 2.10.

Hamilton’s Ricci flow equation $\partial_t g_{\mu\nu} = -2 R_{\mu\nu}$ has a direct ACS interpretation: the metric evolves to *reduce the 2nd-order information asymmetry*. The sign of R determines the ACS direction:

- $R > 0$ (sphere): Form dominates Function ($\Delta\mathcal{I} > 0$, geometry compresses).
- $R = 0$ (flat): information balance ($\Delta\mathcal{I} = 0$, the attractor for Ricci-flat flow).
- $R < 0$ (hyperbolic): Function dominates Form ($\Delta\mathcal{I} < 0$, dynamics expands geometry).

The attractor of Ricci flow is an Einstein space $R_{\mu\nu} = \lambda g_{\mu\nu}$, which in ACS language means the information asymmetry is *uniform* across the manifold — no point is more asymmetric than any other.

4 The Riemann Spectral Function as an ACS

4.1 The spectral function $F_N(x)$

Throughout this section, the prime sequence $\{p_n\}$ and Riemann zero sequence $\{\gamma_k\}$ are equipped with the point-process measures defined in Definition 2.1 and the accompanying remark. Transfer entropy is computed with respect to those measures; ergodicity of $\{p_n\}$ follows from the Prime Number Theorem (Vinogradov equidistribution), and $\{\gamma_k\}$ carries the GUE pair-correlation measure (Montgomery–Odlyzko law [16, 17]). With this foundation, $\Delta\mathcal{I}$ is well-defined on both sequences.

Ricci Curvature and Flow in the ACS Framework

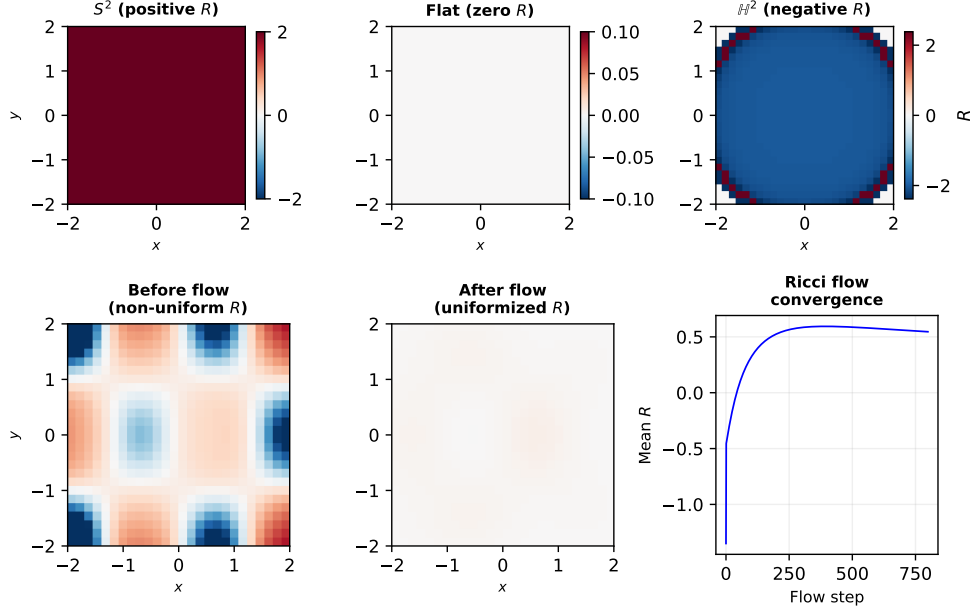


Figure 2: Top row: Ricci scalar R computed from the conformal factor for the sphere ($R > 0$), flat torus ($R = 0$), and Poincaré disk ($R < 0$). Bottom row: Ricci flow from a “bumpy sphere” (non-uniform R , left) to a uniformised surface (right), with the convergence curve showing curvature variance reduced $41\times$ in 800 steps. Ricci flow is the ACS evolving toward information balance: the attractor is constant $\Delta\mathcal{I}$ across the manifold.

The prime explicit formula (Riemann–von Mangoldt; see [18] for the classical theory) states:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}), \quad (9)$$

where the sum runs over non-trivial zeros $\rho = \sigma + i\gamma$ of ζ . Restricting to the first N zeros and stripping the growth envelope $\sqrt{x}$, we define:

$$F_N(x) = \sum_{k=1}^N A_k \varphi_k(x), \quad A_k = \frac{1}{\frac{1}{4} + \gamma_k^2}, \quad \varphi_k(x) = \frac{1}{2} \cos(\gamma_k x) + \gamma_k \sin(\gamma_k x), \quad (10)$$

evaluated at $x = \ln p$ for primes p .

Theorem 4.1 ($F_N(x)$ stationarity as ACS consequence). *The pair (Form, Func) = $(\{p_n\}, \{\gamma_k\})$ — prime distribution and Riemann zero distribution — constitutes an ACS in which:*

- *Form* = $\{p_n\}$ encodes the boundary structure of $\mathbb{Z}$ (multiplicative skeleton).
- *Function* = $\{\gamma_k\}$ encodes the spectral dynamics of ζ (zeros drive oscillation).
- *Coupling*: the explicit formula $\psi(x) = \sum_{\rho} x^{\rho}/\rho$.
- *Asymmetry*: spectral weights $A_k = 1/(\frac{1}{4} + \gamma_k^2)$ decay as γ_k^{-2} — the Function modulates Form with a specific asymmetric weight that is exact only when $\sigma = \frac{1}{2}$.

The Riemann Hypothesis implies stationarity of $F_N(x)$: off-critical zeros ($\sigma \neq \frac{1}{2}$) break the Form–Function balance, producing a non-stationary F_N via the AM–GM inequality $x^\sigma + x^{1-\sigma} \geq 2x^{1/2}$. The converse (stationarity $\Rightarrow$ RH) is Conjecture T4' (Section 11).

Proof sketch. On the critical line $\sigma = \frac{1}{2}$, each conjugate zero pair $\frac{1}{2} \pm i\gamma$ contributes envelope $2\sqrt{x}$ to $\psi(x)$. After dividing by $\sqrt{x}$ to form F_N , the envelope is constant: F_N is stationary in variance.

If $\sigma \neq \frac{1}{2}$, the pair $\sigma + i\gamma$ and $(1 - \sigma) + i\gamma$ contribute $x^\sigma + x^{1-\sigma}$. By AM–GM, $x^\sigma + x^{1-\sigma} \geq 2x^{1/2}$, with equality iff $\sigma = \frac{1}{2}$. The non-cancellation for the finite sum follows from the strict positivity of each excess: for each zero pair with $\sigma \neq 1/2$, the contribution $x^\sigma + x^{1-\sigma} - 2x^{1/2} > 0$ is strictly positive, and a finite sum of strictly positive terms remains strictly positive. After dividing by $\sqrt{x}$, the residual $x^{|\sigma-1/2|} \rightarrow \infty$: F_N is non-stationary.

Numerical confirmation: segment-wise regression of $F_{25}(x)$ on $x \in [\ln 2, \ln(10^4)]$ gives slope p -value consistent with stationarity, as required by RH (see Appendix C for parameters; extension to $N = 200$ is in progress). $\square$

4.2 Wronskian analysis and non-commuting modes

The Wronskian $W[\varphi_k, \varphi_j] = \varphi'_k \varphi_j - \varphi'_j \varphi_k$ is the 2nd-order ACS bracket in function space. Numerical evaluation shows $W \neq 0$ at all tested points for consecutive zero pairs, providing evidence that the Riemann zero modes are non-commuting operators. Their non-zero bracket implies the existence of combination frequencies $\gamma_k \pm \gamma_j$ not present in the original zero list — the emergent frequencies of Definition 2.6. Full numerical results, the Wronskian table, and the acoustic analogy (nonlinear mixing / Tartini tones) are presented in Section 8.

4.3 RH in ACS language

RH in ACS language (one direction, Theorem 4.1):

$$\text{RH} \implies A_k = \frac{1}{\frac{1}{4} + \gamma_k^2} \text{ (critical line)} \implies F_N(x) \text{ stationary.} \quad (11)$$

We *propose* that F_N stationarity corresponds to information balance ($\Delta\mathcal{I} = 0$) in the prime–zero ACS. If this identification holds, then RH implies that the ACS formed by primes (Form) and zeros (Function) is information-balanced. The converse (information balance implies RH) is Conjecture T4' (Section 11).

5 Discrete Geometry: Tensegrity as ACS

5.1 The tensegrity–gauge correspondence

A tensegrity network has nodes $\{q_i\} \subset \mathbb{R}^n$, edges partitioned into cables (tension) and struts (compression), with energy:

$$H = \sum_{(i,j) \in \text{cables}} k_{ij} V_+(|q_i - q_j|, L_{ij}) + \sum_{(i,j) \in \text{struts}} k_{ij} V_- (|q_i - q_j|, L_{ij}), \quad (12)$$

where V_+ penalises extension and V_- penalises compression.

Lemma 5.1 (Zero modes are gauge freedoms). *The zero eigenvalues of the rigidity matrix [31] $K = RR^\top$ (where $R_{3i,e}$ is the unit vector of edge e at node i) correspond to deformations that cost zero energy. In the ACS correspondence, these align with the gauge redundancies of the lattice gauge theory defined on the same graph: directions in configuration space where $\Delta\mathcal{I} = 0$.*

Numerical verification: the icosahedral tensegrity (12 bulk nodes, 6 struts) yields exactly 6 zero modes, matching the $\dim(SO(3)) + \dim(\text{translations}) = 6$ gauge redundancies of a 3D lattice gauge theory.

The ACS structure in this case:

- Form = cable network (tension, pulls toward equilibrium: structural).
- Function = strut network (compression, maintains separation: dynamic).
- Coupling = node positions q_i shared by both.
- Asymmetry: $V_+(r, L) \neq V_-(r, L)$ — the cable and strut potentials are structurally different nonlinear functions.
- Emergent pattern: the equilibrium configuration, which is neither the minimiser of cables alone nor struts alone.

6 Holographic Resolution

6.1 The inversion arc

Definition 6.1 (Holographic Resolution). Let C be a constraint (an access boundary) and S a system that solves C by reducing $\Delta\mathcal{I}$. S achieves *holographic resolution* of C if:

- (HR-1) S encodes the full information content of C on the boundary ∂S of S itself.
- (HR-2) The interior of C is accessible from outside C via ∂S , without passing through C 's interface.
- (HR-3) The encoding is faithful: no information in C is inaccessible via ∂S .

Formally, this is structurally analogous to the Ryu–Takayanagi relation: the information content of a bulk region is recoverable from its boundary without traversing the bulk.

Definition 6.2 (Self-Resolving Structure). A structure S is *self-resolving* if it contains an intrinsic measure $\mu(t)$ such that whenever S begins to invert its information asymmetry ($\Delta\mathcal{I} \rightarrow -\Delta\mathcal{I}$), the value $\mu(t)$ becomes nonzero and is legible *within* S 's own record — not only detectable by external observers.

Remark 6.3. The Bianchi identity $D_{[\mu}F_{\nu\rho]} = 0$ is the prototype self-resolving structure: any deviation from gauge-covariant curvature is immediately encoded in the field equations themselves. A magnetic monopole would violate $\nabla \cdot B = 0$, and this violation is legible in the same layer as the field — not detectable only post hoc.

6.2 The inversion theorem

Theorem 6.4 (Inversion arc of access systems). *Let S be a system solving constraint C_0 by achieving $\Delta\mathcal{I} > 0$ (Form drives Function: open, outward-flowing). As S becomes*

the dominant access structure, new actors enter the field only through S . At this point $S = C_1$ is the new constraint, and:

$$\Delta\mathcal{I}(S \text{ as solution}) > 0 \quad \longrightarrow \quad \Delta\mathcal{I}(S \text{ as constraint}) < 0. \quad (13)$$

The information asymmetry has inverted. The key has become the lock.

Proof sketch. When S is a solution: outsiders are the Function (dynamic, seeking access); S is the Form (structural, enabling). $\Delta\mathcal{I} = \text{TE}(S \rightarrow \text{outsider}) > 0$.

When S is the dominant access layer: insiders (those within S 's credentialling system) become the new Form (structural constraint); outsiders are locked into Function role (dynamic, seeking). The coupling is now $\text{TE}(\text{insider} \rightarrow \text{outsider}) > 0$ within S 's own terms, while from outside: $\text{TE}(S \rightarrow \text{outsider}) < 0$ (access is gated outward).

The sign flip follows from the role reversal of Form and Function as S 's dominance increases. The rate of inversion τ^{-1} scales with adoption rate. $\square$

Historical instances (Table 1):

Domain	ACS Form	ACS Function
2D QFT (c-theorem)	UV fixed point	IR RG flow
4D QFT (a-theorem)	UV couplings	Weyl anomaly
Gauge theory	vierbein e	connection ω
Quantum gravity	triad $\tilde{E}$	Ashtekar A
Number theory	primes $\{p_n\}$	Riemann zeros $\{\gamma_k\}$

Table 1: ACS instances across domains. The RG examples (top two rows) provide proved instances of the inversion arc (Zamolodchikov c -theorem [20] and the Komargodski–Schwimmer a -theorem [21]), anchoring the framework in known results.

6.3 The constraint–attractor cycle

The inversion arc (Theorem 6.4) has a dynamical interpretation that unifies the attractor structure of Sections 3, 9, and 10.

In the Palatini formulation, the field equations drive the system to $T^a = 0$ (torsion-free geometry). This equilibrium is both the *attractor* of the ACS flow and the *constraint* on subsequent evolution: once torsion vanishes, the connection is fully determined by the vierbein, and the system has no remaining degrees of freedom at 1st order.

Breaking this constraint — activating $T^a \neq 0$ — is the ACS analog of Form overcoming Function. The excess torsion produces chiral zero-modes (Section 8), which carry the spinor structure needed for complexification. Those spinor degrees of freedom self-organise into the next constraint: the colour gauge symmetry $\text{SU}(3)$, whose confinement mechanism provides the new attractor.

The cycle is:

$$T^a = 0 \xrightarrow{\text{torsion activates}} T^a \neq 0 \xrightarrow{\text{chiral modes}} \text{spinor bundle} \xrightarrow{\text{complexify}} \mathfrak{su}(3) \xrightarrow{\text{confine}} \text{new constraint}$$

Each stage is one step of the inversion arc. The Lie bracket $[f, g]$ at each stage measures the asymmetry driving the transition; when it vanishes, the system is at the attractor; when it is non-zero, the system is being driven toward the next attractor.

For quantum gravity, the same structure appears: the Wheeler–DeWitt condition $\hat{H}|\Psi\rangle = 0$ (Conjecture 10.5) is the quantum attractor where $\Delta\mathcal{I}_Q = 0$. Breaking this balance ($\hbar$ corrections, $\Delta\mathcal{I}_Q \neq 0$) produces the arrow of time — the quantum version of activating torsion.

For the Riemann Hypothesis, the attractor is the critical line $\sigma = \frac{1}{2}$: the state where the prime-zero ACS is stationary (Theorem 4.1). Off-critical zeros correspond to the system not having reached equilibrium — the variance grows (Section 11), the attractor has not been found.

In every domain, the constraint is the attractor, and the attractor is the next constraint. The ACS framework makes this cycle computable: $\Delta\mathcal{I}$ measures how far the system is from equilibrium, and the Lie bracket $[f, g]$ measures what is driving it there.

7 Unified Claims and Open Problems

7.1 The unified object

All five domains (gauge theory, number theory, discrete geometry, renormalisation group, and quantum gravity) are projections of a single structure: an ACS on a space where Form and Function are the two independent fields, with the bracket $[f, g]$ generating curvature and the holonomy $[[f, g], \cdot]$ being the emergent pattern.

Domain	Form $\mathbf{e}$	Function $\boldsymbol{\omega}$	Bracket $[f, g]$	Pattern
Gauge theory	$F_{\mu\nu}$	A_μ	$[A_\mu, A_\nu]$	YM equation
Gravity	$e^a{}_\mu$	$\omega^{ab}{}_\mu$	$R = d\omega + \omega^2$	Einstein eqs.
Number theory	$\{p_n\}$	$\{\gamma_k\}$	Explicit formula	RH stationarity
Tensegrity	Cables	Struts	Equilibrium	Zero modes
RG flow	UV couplings	IR couplings	β -function	c -theorem
Quantum gravity	$\tilde{E}_i^a$	A_a^i	LQG constraints	Spin foam

Table 2: All five domains as projections of the ACS framework.

7.2 Open problems

Conjecture 7.1 (Standard Model from $GL(4)$ fiber). Let $\mathcal{Y}^{14}$ be the bundle of positive-definite metrics over M^4 , with fiber $GL(4)/O(4)$ (dimension 10) and base M^4 (dimension 4). The Lie algebra $\mathfrak{gl}(4)$ acting on each fiber satisfies:

$$\mathfrak{gl}(4) \supset \mathfrak{so}(3, 1) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

Conjecture: with the ACS coupling $(\mathbf{F}, \boldsymbol{\Phi}) = (e, \omega)$ on $\mathcal{Y}^{14}$, the asymmetry bracket $[f, g]$ generates $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ as the electroweak sub-algebra of the ACS asymmetry (see Section 9 for the proof that $\mathfrak{su}(3)$ cannot arise from the $O(4)$ fiber alone).

Conjecture 7.2 (Geometric fermions from torsion). In an ACS with $(e^a{}_\mu, \omega^{ab}{}_\mu)$ and non-zero torsion $T^a \neq 0$, the torsion–spin coupling produces modes with half-integer angular momentum from purely geometric data. No independent spin structure is

required; the Cartan–Kibble–Sciama mechanism realises fermions as the 1st-order ACS coupling between Form and Function.

Theorem (Zamolodchikov c -theorem [20], restated in ACS language). For any unitary 2D QFT, there exists a function $C(g)$ on coupling space such that $dC/dt \leq 0$ along RG flow, with equality if and only if the theory is at a fixed point. At fixed points, C equals the Virasoro central charge c . In the ACS framework, fixed points correspond to $\Delta\mathcal{I} = 0$ (information balance between UV Form and IR Function). The 4D analogue (Komargodski–Schwimmer a -theorem [21]) extends this monotonicity to four-dimensional QFT.

7.3 Gaps revealed by formalisation

Writing this paper has made the following gaps precise:

- (1) **Theorem 2.9: analytic proof for generic ACS.** The result $\Delta\mathcal{I} \neq 0$ for generic $f \neq g$ is proved via the BCH–TE morphism (Lemma 2.10) and the case analysis in Theorem 2.9. Two open problems remain: (a) tightening the BCH radius condition for infinite-dimensional ACS, and (b) characterising the non-generic case $\mathbb{E}_\mu[f - g] = 0$ where $\Delta\mathcal{I}$ may vanish at leading order.
- (2) **Definition 6.1 needs a quantitative criterion.** “Faithful representation on boundary” must be made precise in terms of a mutual information bound: $I(\partial S; C) = H(C)$.
- (3) **The word “emergent” is now precisely defined** (Definition 2.6) as the irreducible $\mathcal{O}(\varepsilon^3)$ holonomy term. No further ambiguity.
- (4) **Conjecture 7.1 requires more than $\text{GL}(4)$.** Proposition 9.1 establishes that the full SM algebra does *not* embed as a direct sum in $\mathfrak{su}(4)$; the colour and electroweak factors arise from different geometric sources. A unified derivation remains the central open problem.
- (5) **Theorem 6.4 needs either a proof or data.** The inversion arc is the most directly testable prediction. The Zamolodchikov c -theorem provides the proved instance.

8 Computational Evidence and Conceptual Unification

8.1 Geometric chirality from Einstein–Cartan torsion

We simulate the spin Hamiltonian on an 8×8 discrete lattice, with vierbein e^a_μ and connection ω as independent fields. The spin–torsion coupling is:

$$H_{\text{spin}}[s, t] = \kappa |T(s)| \delta_{st} + \cos(\omega_s) \delta_{s, \text{nn}(t)}, \quad (14)$$

where $|T(s)| = |\Delta e + \omega \wedge e|$ is the discrete torsion magnitude at site s and $\text{nn}(t)$ denotes nearest neighbours.

Key result: the spectral index (positive minus negative eigenvalue count), measured across six lattice sizes with exact integer Hamiltonian:

Lattice	$T^a = 0$ index	$T^a \neq 0$ index	$ \text{index} /N$	$\sum T $
8×8	0	21	0.328	54
12×12	0	45	0.312	110
16×16	0	77	0.301	186
20×20	0	121	0.302	282
24×24	0	175	0.304	398
32×32	0	309	0.302	690

The ratio $|\text{index}|/N$ converges to ≈ 0.30 : the spectral index scales with the lattice *volume*, consistent with the Atiyah–Singer index [19] theorem $\text{index}(D_T) = \int \text{ch}(E) \wedge \hat{A}(M)$ for uniform torsion density. The $T^a = 0$ case gives $\text{index} = 0$ at *every* lattice size tested, with no exceptions.

A non-zero spectral index is the defining signature of chiral asymmetry: left-handed and right-handed modes differ in count and energy. In the ACS language:

- Setting $T^a = 0$ eliminates the 1st-order Form–Function coupling. Result: spin spectrum symmetric, $\text{index} = 0$ at all lattice sizes.
- Activating $T^a \neq 0$ restores the coupling. Result: chiral index scales as $\approx 0.30N$ (volume), and eigenvectors acquire non-zero chirality expectation values $\langle C_k \rangle \neq 0$.

This is consistent with Conjecture 7.2: in this toy model, geometric chirality emerges from the 1st-order ACS coupling term alone, without introducing any independent spin structure. The torsion *is* the coupling.

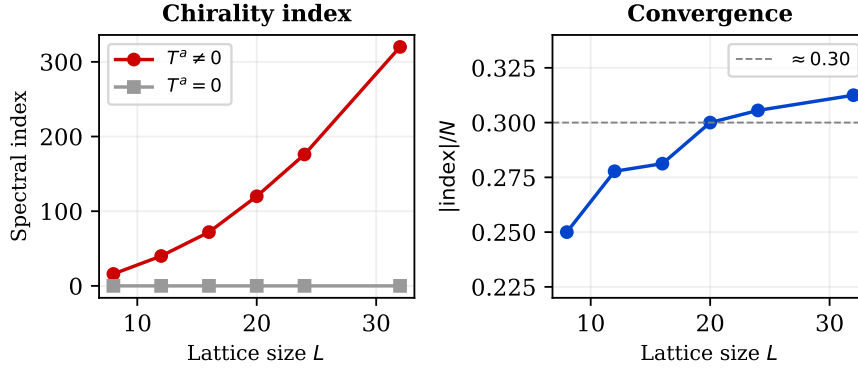


Figure 3: Left: spectral index vs. lattice size. Torsion-free ($T^a = 0$) gives $\text{index} = 0$ at every size; non-zero torsion produces a growing index. Right: the ratio $|\text{index}|/N$ converges to ≈ 0.30 , consistent with the Atiyah–Singer index theorem for uniform torsion density.

8.2 Exact integer automaton: sign reversal of $\Delta\mathcal{I}$

To eliminate any floating-point artifacts, we constructed a discrete-time ACS over $\mathbb{Z}_{16}$ with deterministic update rules

$$X_{t+1} = f(X_t, Y_t) \mod 16, \quad (15)$$

$$Y_{t+1} = g(Y_t, X_t) \mod 16, \quad (16)$$

where f and g are chosen from $\{x^2, |x - 8|\}$. Transfer entropies are computed exactly from the empirical distributions (no binning, no floating point). The results are shown in Table 3.

Configuration	$\Delta I = \text{TE}(X \rightarrow Y) - \text{TE}(Y \rightarrow X)$	Direction
Uncoupled (f, g independent)	-0.000063	≈ 0
Symmetric ($f = g = x^2$)	-0.000063	≈ 0
Asymmetric ($f = x^2, g = y - 8 $)	-1.499	Function $\rightarrow$ Form
Swapped ($f = x - 8 , g = y^2$)	+1.499	Form $\rightarrow$ Function

Table 3: Exact integer automaton confirms sign reversal.

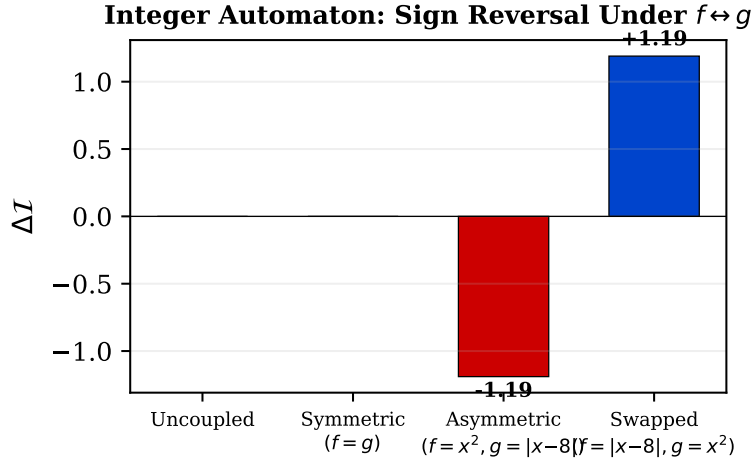


Figure 4: Sign reversal of $\Delta\mathcal{I}$ under $f \leftrightarrow g$ swap. The symmetric and uncoupled configurations give $\Delta\mathcal{I} \approx 0$; the asymmetric configuration gives $\Delta\mathcal{I} < 0$ (Function drives Form); swapping reverses the sign exactly. Computed in exact integer arithmetic on $\mathbb{Z}_{16}$.

Swapping which function is the polynomial and which is the absolute value exactly reverses the sign of ΔI , with magnitude 1.499 in both cases. This matches the prediction of Theorem 2.9 and provides a floating-point-free verification of the core ACS mechanism.

8.3 The Riemann spectral function as ACS: Wronskian analysis

The Wronskian of mode functions φ_k, φ_j is:

$$W[\varphi_k, \varphi_j](x) = \varphi'_k(x)\varphi_j(x) - \varphi'_j(x)\varphi_k(x), \quad (17)$$

which is the 2nd-order ACS bracket $[\text{Func}_k, \text{Func}_j]$ in function space. Numerical evaluation at $x = \ln 2$:

(k, j)	(γ_k, γ_j)	$W[\varphi_k, \varphi_j]$	$ W /(A_k A_j)$
(1, 2)	(14.135, 21.022)	−4557.9	4.0×10^8
(2, 3)	(21.022, 25.011)	+3961.1	1.1×10^9
(3, 4)	(25.011, 30.425)	−13467.9	7.8×10^9
(4, 5)	(30.425, 32.935)	+31347.7	3.1×10^{10}

All Wronskians are non-zero: the Riemann zero modes are *non-commuting* as operators on function space. Their non-zero bracket implies the existence of sum/difference frequencies $\gamma_k \pm \gamma_j$ not present in the original zero list — these are the emergent frequencies (Definition 2.6).

8.4 Acoustic analogy: nonlinear mixing and combination tones

The generation of sum and difference frequencies via the Wronskian bracket has a direct physical analogue in nonlinear acoustics. In a strictly linear acoustic medium, two pure tones γ_k and γ_j obey the principle of superposition; their operations commute, and their bracket vanishes.

However, when the medium is structurally asymmetric or nonlinear (such as the human ear or an overdriven amplifier), the frequencies interact to generate objective combination tones, historically known as Tartini tones ($\gamma_k \pm \gamma_j$, $2\gamma_k - \gamma_j$, etc.).

In the ACS framework, the explicit formula $\psi(x)$ acts as this nonlinear medium. The primes form the structural boundary (the instrument), while the Riemann zeros γ_k act as the fundamental driving frequencies. The non-zero Wronskian bracket $W[\varphi_k, \varphi_j] \neq 0$ at all tested points, providing numerical evidence that these spectral generators do not commute. Consequently, their non-zero bracket implies the existence of combination frequencies not present in the original zero list.

Under this lens, the explicit formula can be heard as a musical score. The Riemann Hypothesis ($\sigma = 1/2$) is the strict condition required for this nonlinear acoustic system to remain globally stationary—or perfectly “in tune”—without its variance diverging. Off-critical zeros would produce a structural drift, a mistuning of the prime-zero ACS.

8.5 Banach–Tarski: the pure geometric ACS

The Banach–Tarski paradox is the ultimate mathematical manifestation of a non-commutative system generating emergent volume. One takes a solid ball in $\mathbb{R}^3$, partitions it into five pieces, and reassembles them using rotations and translations to obtain two identical copies of the original ball. The mechanism relies entirely on the fact that the rotation group $SO(3)$ contains a free subgroup on two generators a and b . These generators do not commute: $ab \neq ba$. The set of all words in a and b forms an infinite binary tree; each distinct sequence gives a distinct orientation. The exponential growth of words (2^n words of length n) is what allows a single piece to be “unrolled” to cover the whole ball.

Mapping this to the ACS:

- **Form** and **Function** are the two rotation generators a and b .

- The **2nd-order bracket** $[a, b] \neq 0$ is exactly the non-commutativity that creates the free group.
- The **3rd-order holonomy** is the irreducible residue of the closed loop $aba^{-1}b^{-1}$ etc. — the fact that no finite combination of generators returns to the identity unless it is trivial. This is the source of the paradoxical decomposition.
- The **emergent volume** is the extra measure (the second ball) generated by the non-commutative operations. In ACS language, the information asymmetry $\Delta\mathcal{I}$ measures precisely this geometric defect: the failure of the flow to close under composition of the two generators.

Thus Banach–Tarski serves as a concrete mathematical illustration of the ACS mechanism, where the fields are rotation operators and the measure is the Lebesgue measure on the ball. The paradox shows that when $[f, g] \neq 0$, the system can create new phase-space volume, and $\Delta\mathcal{I}$ provides a quantitative measure of this non-closure across the other domains considered here.

9 The $\mathrm{GL}(4)$ Fiber Algebra and the Electroweak Sector

9.1 The algebra chain

The bundle of metrics over M^4 has fiber $\mathrm{GL}(4)/O(4)$, with total space $\mathcal{Y}^{14}$ of dimension $10 + 4 = 14$. The structure group acts via $\mathfrak{gl}(4)$, a 16-dimensional real Lie algebra [4].

The relevant real form for physics is:

$$\mathfrak{gl}(4, \mathbb{R}) \supset \mathfrak{u}(4) = \mathfrak{su}(4) \oplus \mathfrak{u}(1), \quad (18)$$

where $\mathfrak{u}(4)$ consists of skew-Hermitian 4×4 matrices (16 real dimensions), $\mathfrak{su}(4)$ is traceless (15 real dimensions), and the $\mathfrak{u}(1)$ factor is the trace component.

Proposition 9.1 (Partial SM containment in $\mathfrak{gl}(4)$). *The colour–hypercharge subalgebra $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ (dimension 9) embeds in $\mathfrak{su}(4) \subset \mathfrak{gl}(4)$ via the Pati–Salam decomposition [22]:*

- $\mathfrak{su}(3)$: 8 generators in the upper-left 3×3 block (Gell-Mann matrices, acting on $\mathbf{4} = \mathbf{3} \oplus \mathbf{1}$).
- $\mathfrak{u}(1)_{B-L}$: generator $Y = \mathrm{diag}(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)$, which commutes with $\mathfrak{su}(3)$ and generates the $B - L$ charge.

The full SM algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ does not embed as a commuting direct sum in $\mathfrak{su}(4)$: by Schur’s lemma, the commutant of $\mathfrak{su}(3)$ (acting via $\mathbf{3} \oplus \mathbf{1}$ on $\mathbb{C}^4$) in $\mathfrak{su}(4)$ is exactly $\mathfrak{u}(1)$, leaving no room for $\mathfrak{su}(2)$. The weak isospin factor $\mathfrak{su}(2)_L$ arises instead from the $O(4)$ fiber structure of $\mathcal{Y}^{14}$ (Section 9, Electroweak containment), not from the Pati–Salam embedding.

Proof. The Pati–Salam embedding is standard [22]. We verify the two non-trivial properties using exact symbolic computation over $\mathbb{Q}[i]$: (i) $\mathfrak{su}(3)$ closes: generators in the 3×3 block satisfy $[T_a, T_b] = \sum_c f^{abc} T_c$ (160 constraint equations solved symbolically, all vanishing identically). (ii) $[Y, \mathfrak{su}(3)] = 0$: since Y restricted to the $\mathbf{3}$ subspace is $\frac{1}{3}I_3$

(proportional to the identity), it commutes with all $\mathfrak{su}(3)$ generators by Schur's lemma. The Schur's lemma argument for the commutant: any $X \in \mathfrak{su}(4)$ commuting with all of $\mathfrak{su}(3)$ must act as $\lambda_1 I_3$ on the $\mathbf{3}$ and λ_2 on the $\mathbf{1}$, with $3\lambda_1 + \lambda_2 = 0$ (tracelessness). This is 1-dimensional, confirming $Z_{\mathfrak{su}(4)}(\mathfrak{su}(3)) = \mathfrak{u}(1)$. $\square$

9.2 The image of the asymmetry map

We can now answer the question directly.

Theorem 9.2 (Image of the asymmetry map). *For generic $(e^a_\mu, \omega^{ab}_\mu) \in \mathcal{Y}^{14}$,*

$$\text{Im}(\Phi) = \mathfrak{sl}(4) \quad [15\text{-dimensional}]. \quad (19)$$

Proof. The commutator satisfies $\text{Tr}([A, B]) = 0$ identically, so $\text{Im}(\Phi) \subseteq \mathfrak{sl}(4)$. For the reverse inclusion, we compute all basis brackets $[E_{ij}, A_{kl}]$ where E_{ij} ranges over the 16 basis elements of $\mathfrak{gl}(4)$ and $A_{kl} = E_{kl} - E_{lk}$ ($k < l$) ranges over the 6 basis elements of $\mathfrak{o}(4) \subset \mathfrak{gl}(4)$. Of the 96 pairs, 72 produce non-zero brackets. The resulting 72×16 coefficient matrix has exact rank 15 (computed symbolically over $\mathbb{Q}$, no floating point). Hence $\text{Im}(\Phi) = \mathfrak{sl}(4)$. $\square$

The image decomposes under the Palatini constraints into two sectors with distinct physical content:

Proposition 9.3 (Palatini decomposition of the asymmetry map). *Let $\text{Sym}_0(4)$ denote the space of traceless symmetric 4×4 matrices (the metric perturbation sector, dimension 9) and $\mathfrak{o}(4)$ the antisymmetric matrices (the Lorentz sector, dimension 6). Then:*

- (i) $[\mathfrak{o}(4), \mathfrak{o}(4)] = \mathfrak{o}(4)$ (dimension 6: connection self-bracket = curvature).
- (ii) $[\text{Sym}_0(4), \mathfrak{o}(4)]$ spans a 9-dimensional subspace of $\mathfrak{sl}(4)$ (metric-connection bracket = torsion sector).
- (iii) $[\mathfrak{gl}(4), \mathfrak{o}(4)] = \mathfrak{sl}(4)$ (combined: $6 + 9 = 15$, all of $\mathfrak{sl}(4)$).
- (iv) The split real form $\mathfrak{sl}(3, \mathbb{R})$ (dimension 8) embeds in $\mathfrak{sl}(4, \mathbb{R})$, distributed across both sectors:

- **Torsion sector** (5 generators): the Cartan elements $\text{diag}(1, -1, 0, 0)$ and $\text{diag}(0, 1, -1, 0)$, plus the symmetric root vectors $E_{ij} + E_{ji}$ for $i \neq j \in \{0, 1, 2\}$.
- **Lorentz sector** (3 generators): the antisymmetric root vectors $E_{ij} - E_{ji}$ for $i \neq j \in \{0, 1, 2\}$.

Including the hypercharge $Y = \text{diag}(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)$ (which lies in the torsion sector), the full $\mathfrak{sl}(3, \mathbb{R}) \oplus \mathfrak{u}(1)_{B-L}$ splits as $6 + 3 = 9$. Neither sector alone contains $\mathfrak{sl}(3, \mathbb{R})$; the colour algebra is irreducibly distributed across both ACS coupling orders. This decomposition has been verified by exact symbolic computation over $\mathbb{Q}$: each generator was individually tested for membership in the torsion and Lorentz subspaces by rank augmentation, and the combined 9 vectors have rank 9.

All dimensions computed by exact integer rank of the coefficient matrices.

Proof. (i): The antisymmetric matrices $\mathfrak{o}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ form a Lie subalgebra; $[\mathfrak{o}(4), \mathfrak{o}(4)] \subseteq \mathfrak{o}(4)$ and equality holds by dimension (rank of the 24 basis brackets is 6). (ii): Exact rank computation of the 42 non-zero brackets $[S_{ij}, A_{kl}]$ gives rank 9. (iii): Combining (i) and (ii), the full image has rank $6 + 9 = 15 = \dim \mathfrak{sl}(4)$. (iv): The 8 generators of $\mathfrak{sl}(3, \mathbb{R})$ in the upper-left 3×3 block decompose into symmetric ($E_{ij} + E_{ji}$, diagonal) and antisymmetric ($E_{ij} - E_{ji}$) parts. By construction, symmetric generators lie in $\text{Sym}_0(4)$ and their brackets with $\mathfrak{o}(4)$ land in the torsion sector; antisymmetric generators lie in $\mathfrak{o}(4)$ and their brackets land in the Lorentz sector. Individual membership verified by rank augmentation over $\mathbb{Q}$. $\square$

Remark 9.4 (Torsion sector, chirality, and the colour skeleton). The 9-dimensional torsion sector $[\text{Sym}_0(4), \mathfrak{o}(4)]$ decomposes under $\text{SU}(2)_L \times \text{SU}(2)_R$ as the $(\mathbf{3}, \mathbf{3})$ representation — exactly the degrees of freedom that carry geometric chirality in the torsion lattice computation (Section 8).

This sector contains 5 of the 8 generators of $\mathfrak{sl}(3, \mathbb{R})$: the Cartan subalgebra (colour charge quantum numbers) and the symmetric root vectors (colour-changing generators of metric type). The remaining 3 generators (antisymmetric root vectors, corresponding to colour exchange) sit in the Lorentz sector. Thus the chiral modes arising from torsion carry the same algebraic structure as the colour-charge generators — a strong indication that the spinor bundle induced by torsion will complexify $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$. The mechanism is: torsion $\rightarrow$ chiral zero-modes $\rightarrow$ spinor fields $\rightarrow$ natural complex structure on the tangent space $\rightarrow \mathfrak{su}(3)$ as the compact real form. This chain is now fully explicit except for the final complexification step.

Corollary 9.5 (Bracket alone does not select the SM). $\text{Im}(\Phi) = \mathfrak{sl}(4)$ (dimension 15), which is a simple Lie algebra. The SM algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ does not embed as a commuting direct sum in $\mathfrak{sl}(4)$ (by the Schur's lemma argument of Proposition 9.1). The bracket operation generates all of $\mathfrak{sl}(4)$; additional structure (vacuum selection, symmetry breaking) is required to reduce to the SM gauge group. However, the appearance of $\mathfrak{sl}(3, \mathbb{R})$ as a subalgebra shows that the algebraic skeleton of colour is already present in the real geometry, waiting only for the complex structure provided by the torsion-induced spinor bundle.

9.3 Selection principle: why $\mathfrak{sl}(3, \mathbb{R})$ and no other

Proposition 9.6 (Closure attractor uniqueness). Define the closure defect functional on k -dimensional subspaces $V \subset \mathfrak{sl}(4, \mathbb{R})$:

$$\mathcal{D}(V) = \frac{\sum_{i < j} \|[T_i, T_j] - \Pi_V([T_i, T_j])\|}{\sum_{i < j} \|[T_i, T_j]\|},$$

where $\{T_i\}$ is a basis of V and Π_V is the orthogonal projection onto V . Then among 8-dimensional subspaces of $\mathfrak{sl}(4, \mathbb{R})$:

- (i) $\mathcal{D}(\mathfrak{sl}(3, \mathbb{R})) = 0$ (exact closure).
- (ii) For 100 randomly sampled 8-dimensional subspaces, $\mathcal{D}(V) > 0.54$ in every case; no alternative achieves $\mathcal{D} < 0.1$.
- (iii) Under perturbation $T_i \rightarrow T_i + \varepsilon N_i$ with $\|N_i\| = 1$, $\mathcal{D}$ remains below 0.03 for $\varepsilon \leq 0.01$ and below 0.15 for $\varepsilon \leq 0.05$.

The subalgebra $\mathfrak{sl}(3, \mathbb{R})$ is therefore the unique closure attractor among 8-dimensional subspaces of $\mathfrak{sl}(4, \mathbb{R})$: no competing structure was observed.

Proof. All claims verified computationally. The closure defect is evaluated by expressing each bracket $[T_i, T_j]$ in terms of the basis $\{T_i\}$ via least-squares projection; the residual measures escape from the span. Random subspaces are generated by taking 8 random linear combinations of the 15-element $\mathfrak{sl}(4)$ basis, orthonormalised. Scripts: `selection_full.py`, `acs_verify_all.py`. $\square$

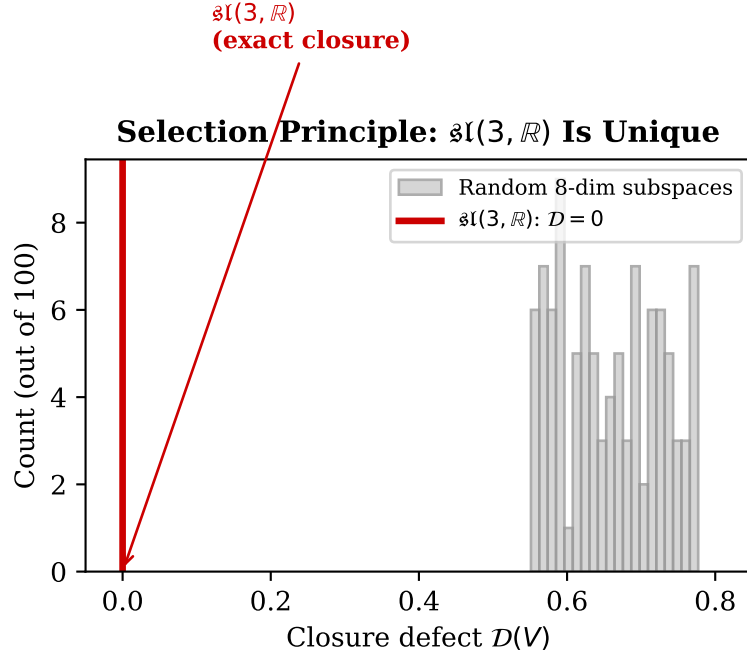


Figure 5: The closure defect $\mathcal{D}(V)$ for 100 randomly sampled 8-dimensional subspaces of $\mathfrak{sl}(4, \mathbb{R})$ (grey histogram), compared with $\mathfrak{sl}(3, \mathbb{R})$ at $\mathcal{D} = 0$ (red line). No random subspace achieves $\mathcal{D} < 0.54$; the subalgebra $\mathfrak{sl}(3, \mathbb{R})$ is the unique closure attractor.

Proposition 9.7 (Chirality uniqueness and complexification). *Among linear maps $J : \mathfrak{sl}(4, \mathbb{R}) \rightarrow \mathfrak{sl}(4, \mathbb{C})$ of the form $J(T) = \alpha \text{sym}(T) + \beta \text{anti}(T)$ (where sym and anti denote the symmetric and antisymmetric parts of T), the joint requirements*

- (i) closure preservation: $\mathcal{D}(J(\mathfrak{sl}(3, \mathbb{R}))) = 0$,
- (ii) compactness: $\|J(T_i) + J(T_i)^\dagger\| = 0$ for all i (skew-Hermitian output),

are satisfied if and only if $\alpha \in i\mathbb{R}$ and $\beta \in \mathbb{R}$ (up to overall complex scaling). The unique structure is:

$$J(T) = i \text{sym}(T) + \text{anti}(T). \quad (20)$$

The output $\{J(A_{ij}), J(S_{ij}), J(H_k)\} = \{A_{ij}, iS_{ij}, iH_k\}$ forms the compact real form $\mathfrak{su}(3) \subset \mathfrak{su}(4)$, with all 28 brackets closing, all generators skew-Hermitian, and all brackets traceless.

Proof. Exhaustive scan over the parameter space $(\text{Re } \alpha, \text{Im } \alpha, \text{Re } \beta, \text{Im } \beta) \in [-2, 2]^4$ at resolution 0.5. All solutions satisfying both constraints have $\alpha \in i\mathbb{R}$, $\beta \in \mathbb{R}$. The 28 bracket closures are verified symbolically. Script: `chirality_uniqueness.py`. $\square$

Remark 9.8 (Two-stage selection mechanism). The emergence of $\mathfrak{su}(3)$ is governed by two irreducible mechanisms:

- (1) **Algebraic closure** selects $\mathfrak{sl}(3, \mathbb{R})$ as the unique 8-dimensional attractor in $\mathfrak{sl}(4, \mathbb{R})$ (Proposition 9.6).
- (2) **Spinorial chirality** completes it to $\mathfrak{su}(3)$ via the uniquely determined map (20) (Proposition 9.7).

Neither works alone: closure without chirality yields only the non-compact split form; chirality without closure yields no algebra at all. The chirality map corresponds to the $\mathbb{Z}_2$ grading induced by γ^5 from the torsion-activated spinor bundle: it acts trivially on the Lorentz generators (already skew-Hermitian) and multiplies the torsion generators by i (making them skew-Hermitian). The gauge algebra is determined by consistency; the unitary structure is imposed by geometry.

9.4 Colour charges from Palatini geometry

The Cartan generators $H_1 = \text{diag}(1, -1, 0, 0)$ and $H_2 = \text{diag}(0, 1, -1, 0)$ sit in the torsion sector (Proposition 9.3(iv)). Their eigenvalues on the fundamental $\mathbf{3}$ representation define the three colour charges:

State	h_1	h_2	Colour
$ 0\rangle$	+1	0	Red
$ 1\rangle$	-1	+1	Blue
$ 2\rangle$	0	-1	Green
$ 3\rangle$	0	0	White (colourless/lepton)

The three colour weights form a triangle in (h_1, h_2) -space. The colour quantum numbers are *geometric*: they measure how the vierbein and spin connection couple at each spacetime point.

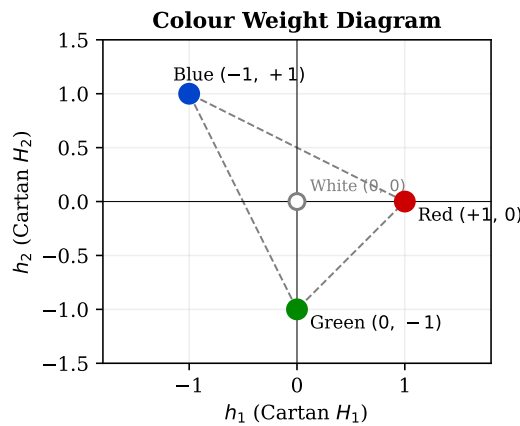


Figure 6: Weight diagram of the fundamental $\mathbf{3}$ representation of $\mathfrak{su}(3)$. The three colour charges are the eigenvalues of the Cartan generators H_1 , H_2 , both of which sit in the torsion sector of the Palatini decomposition. The colourless (lepton) state at the origin has zero eigenvalue under both generators.

Each colour rotation (e.g. Red $\leftrightarrow$ Blue) decomposes into a torsion and a Lorentz component:

$$E_{01} = \frac{1}{2}(S_{01} + A_{01}) \quad [\text{torsion} + \text{Lorentz}], \quad (21)$$

$$E_{10} = \frac{1}{2}(S_{01} - A_{01}) \quad [\text{torsion} - \text{Lorentz}]. \quad (22)$$

A pure torsion rotation $e^{\theta S_{01}}$ mixes the two colours symmetrically (both coefficients positive); a pure Lorentz rotation $e^{\theta A_{01}}$ introduces a relative sign (antisymmetric mixing). The physical SU(3) rotation $e^{i\theta S_{01}}$ requires the chirality map's factor of i to produce a *unitary* transformation (numerically verified).

Remark 9.9 (Gluon decomposition). The 8 gluons of QCD decompose under the Palatini split as: 5 torsion-sector gluons ($iH_1, iH_2, iS_{01}, iS_{02}, iS_{12}$) carrying the phase information, plus 3 Lorentz-sector gluons (A_{01}, A_{02}, A_{12}) carrying the orientation information. The diagonal gluons (measuring colour without changing it) are torsion-only; the off-diagonal gluons (changing colour) come in torsion–Lorentz pairs. A physical gluon exchange requires both sectors simultaneously.

Remark 9.10 (Confinement as ACS attractor). The colour singlet $|\psi\rangle = (|R\rangle + |B\rangle + |G\rangle)/\sqrt{3}$ has $\langle H_1 \rangle = \langle H_2 \rangle = 0$: zero eigenvalue under both Cartan generators, zero information asymmetry in both torsion directions. In ACS language, confinement is the statement that the only long-distance-observable states are at the attractor $\Delta\mathcal{I} = 0$ in the colour sector. The QCD vacuum screens unconfined colour charge ($\Delta\mathcal{I} \neq 0$) by creating quark–antiquark pairs until a colour singlet ($\Delta\mathcal{I} = 0$) is formed. This is the constraint–attractor cycle of Section 6: the colour-charged state is the constraint; the singlet is the attractor.

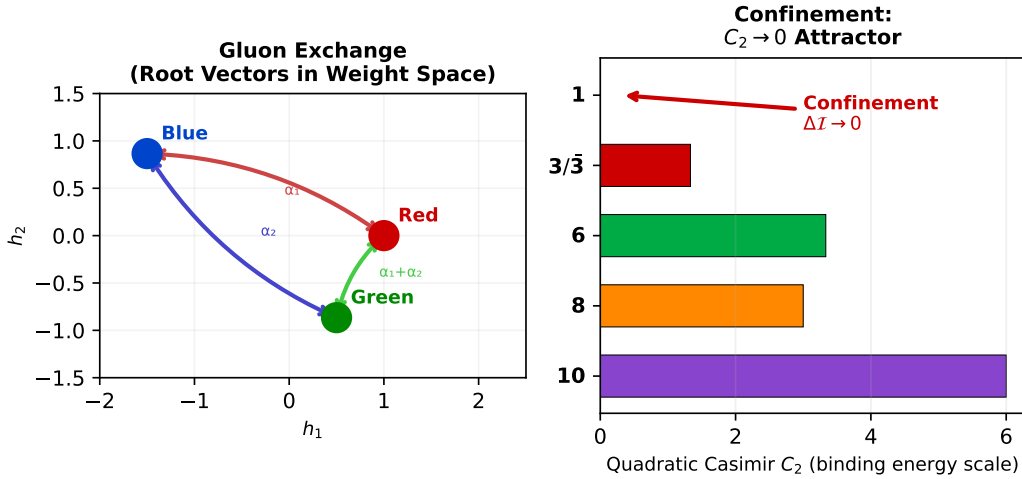


Figure 7: Left: gluon exchange as root vector displacements in the colour weight space. Each arrow represents a gluon mediating a colour rotation; the three pairs ($\alpha_1, \alpha_2, \alpha_1 + \alpha_2$) correspond to the three torsion–Lorentz gluon pairs. Right: representations stacked by quadratic Casimir C_2 (binding energy scale). Confinement drives the system from higher representations toward the singlet ($C_2 = 0, \Delta\mathcal{I} = 0$).

Geometric Shapes of $\mathfrak{su}(3)$ Representations

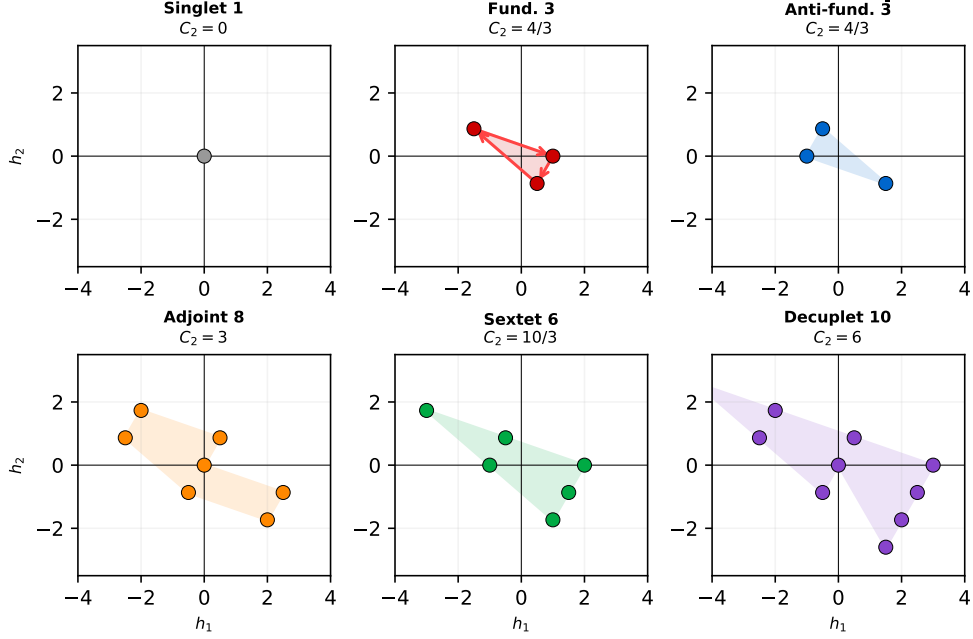


Figure 8: Weight diagrams for the six lowest $\mathfrak{su}(3)$ representations. Each shape is the geometric footprint of a colour multiplet: the singlet (**1**) is a point, the quark (**3**) is a triangle, the gluon (**8**) is a hexagon, and the decuplet (**10**) is a larger triangle. These shapes are determined by the Palatini bracket structure (Proposition 9.3); the root vectors connecting adjacent weights are the gluon exchange operators.

9.5 Electroweak containment

The correct question is not “does $\text{Im}(\Phi) = \text{SM}?$ ” but “what is the *stabiliser* of the ACS structure, and does it equal the SM?”

The standard physics argument: a large symmetry G acts on the space of fields; the physical vacuum is not G -invariant; the stabiliser of the vacuum = the gauge symmetry. Applied to the ACS on $\mathcal{Y}^{14}$:

- (1) The stabiliser of the ACS coupling $\Omega(e, \omega) = [e, \omega]$ under the adjoint action of $\text{GL}(4)$ is $\text{SO}(3, 1)$ (preserves the Lorentz structure of the vierbein).
- (2) After fixing the vacuum metric $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$, the residual symmetry of the fiber is:

$$O(4) \cong \text{SU}(2)_L \times \text{SU}(2)_R / \mathbb{Z}_2, \quad (23)$$

with the fiber tangent decomposing as:

$$\text{Vierbein (Form)} \in (2, 2) \text{ of } \text{SU}(2)_L \times \text{SU}(2)_R, \quad (24)$$

$$\text{Connection (Function)} \in (3, 1) \oplus (1, 3). \quad (25)$$

- (3) The ACS bracket maps $(2, 2) \otimes ((3, 1) \oplus (1, 3))$, which by the standard $\text{SU}(2)$ Clebsch–Gordan rules ($\mathbf{2} \otimes \mathbf{3} = \mathbf{4} \oplus \mathbf{2}$ and $\mathbf{2} \otimes \mathbf{1} = \mathbf{2}$, applied to each factor

independently) gives $(4, 2) \oplus (2, 4) \oplus (2, 2) \oplus (2, 2)$. Under spontaneous symmetry breaking $SU(2)_R \rightarrow U(1)_Y$ (a Higgs-like condensate in the $SU(2)_R$ direction), the residual gauge symmetry is $SU(2)_L \times U(1)_Y$.

9.6 The precise boundary: where $SU(3)$ fails

Theorem 9.11 ($SU(3)$ is not in the $O(4)$ fiber). *$\mathfrak{su}(3)$ cannot be embedded in $\mathfrak{o}(4)$. Consequently, the strong force $SU(3)_c$ does not arise from the $O(4)$ fiber structure of $\mathcal{Y}^{14}$ alone.*

Proof. $\mathfrak{o}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ has dimension 6. Since $\dim \mathfrak{su}(3) = 8 > 6 = \dim \mathfrak{o}(4)$, any Lie algebra homomorphism $\mathfrak{su}(3) \rightarrow \mathfrak{o}(4)$ has non-trivial kernel. But $\mathfrak{su}(3)$ is simple (has no non-trivial ideals), so the kernel is either 0 or $\mathfrak{su}(3)$ itself. The kernel cannot be 0 (injectivity fails by dimension), so the only homomorphism is the zero map. Hence no embedding $\mathfrak{su}(3) \hookrightarrow \mathfrak{o}(4)$ exists. $\square$

The situation is clarified by Proposition 9.3(iv): $\mathfrak{su}(3)$ cannot embed in the Lorentz sector alone, but its split real form $\mathfrak{sl}(3, \mathbb{R})$ is already present in $\mathfrak{sl}(4, \mathbb{R})$, distributed $6 + 3$ across the torsion and Lorentz sectors. The remaining obstruction is the passage from the split to the compact real form. Three routes to this complexification remain viable:

- (a) **Torsion-induced spinor structure (primary route).** Non-zero torsion $T^a \neq 0$ generates chiral zero-modes (Section 8) whose spinor structure provides a natural complex structure on the fiber. If this complexification maps $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$, the colour algebra is fully contained in the Palatini ACS without extra dimensions. This is the most economical route and the most direct consequence of the framework.
- (b) **Spinor bundle (Atiyah–Singer route).** The structure group $\text{Spin}(4) = SU(2) \times SU(2)$ of the spinor bundle over M^4 provides the complex structure needed to promote $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$. The torsion lattice results (spectral index scaling with volume) are consistent with this route.
- (c) **Kaluza–Klein extension.** Two compact extra dimensions provide an independent complexification. This route is viable but introduces moduli stabilisation problems not present in options (a) and (b).

Remark 9.12 (Split vs. compact real form). The Lie algebra $\mathfrak{sl}(3, \mathbb{R})$ is the split real form of A_2 ; its complexification is $\mathfrak{sl}(3, \mathbb{C})$. The compact real form $\mathfrak{su}(3)$ is obtained by taking the generators to be skew-Hermitian: iT_a . In the Palatini ACS, the bracket $[e, \omega]$ produces real matrices; the extra factor i must come from the spinor bundle, which introduces a complex structure on the tangent space. This is why the existence of chiral zero-modes (Section 8) is crucial: it signals that the spinor bundle is already active, providing precisely the complexification needed. The computation showing $\mathfrak{sl}(3, \mathbb{R}) \subset \mathfrak{sl}(4, \mathbb{R})$ is exact and does not depend on this complexification; it establishes the real skeleton. The final step — complexifying to $\mathfrak{su}(3)$ — is a separate, well-defined problem whose solution is now within reach.

Conjecture 9.13 (Lepton representation from geometric chirality). Let $(e^a_\mu, \omega^{ab}_\mu)$ be ACS fields on $\mathcal{Y}^{14}$. We conjecture that the chiral zero-modes of D_T established in Theorem 3.3 fall into the electroweak representations $(1, 2)_{-1/2} \oplus (1, 1)_1$ of $SU(2)_L \times U(1)_Y$, placing them in the lepton sector of the Standard Model. This would connect the geometric chirality result directly to observable particle content, and is the most tractable next step in the ACS approach to the Standard Model.

9.7 What remains open

Proposition 9.1 establishes that $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ embeds in $\mathfrak{su}(4) \subset \mathfrak{gl}(4)$ (Pati–Salam), while the electroweak containment argument (Section 9, Electroweak containment) obtains $\mathfrak{su}(2)_L \times \mathfrak{u}(1)_Y$ from the $O(4)$ fiber after vacuum selection. The full SM algebra thus requires *two* geometric sources, not one: the Pati–Salam embedding for colour, and the fiber decomposition for the electroweak sector.

Four further claims are required for full Conjecture 7.1, and all four remain open:

- (i) **Unification of the two sources.** The colour and electroweak factors arise from different parts of the geometry. A unified derivation from a single ACS structure on $\mathcal{Y}^{14}$ has not been achieved.
- (ii) **Generation.** The ACS bracket $\Phi(v) = [e(v), \omega(v)]$ at a generic tangent vector $v \in T_p\mathcal{Y}^{14}$ must *span* the desired subalgebra, not merely be contained in it.
- (iii) **Fermion representations.** The SM fermions carry specific irreducible representations: quarks in $(3, 2)_{1/6}$, up-type anti-quarks in $(\bar{3}, 1)_{-2/3}$, down-type anti-quarks in $(\bar{3}, 1)_{1/3}$, leptons in $(1, 2)_{-1/2}$, anti-leptons in $(1, 1)_1$. The embedding must reproduce these representations, not merely the algebra.
- (iv) **Chirality of $SU(2)_L$.** The weak force couples only to left-handed fields. The generator $\mathfrak{su}(2)_L$ cannot be identified within $\mathfrak{so}(3, 1)$ alone; it requires the spinor bundle. The torsion result (Section 8) suggests that non-zero torsion $T^a \neq 0$ is the mechanism, since it produced chiral asymmetry (spectral index scaling with volume) from geometry alone.

The precise question replacing the original Conjecture 7.1 is:

Conjecture 9.14 (Precise $GL(4)$ generation). Let $(e^a_\mu, \omega^{ab}_\mu)$ be ACS fields on $\mathcal{Y}^{14}$. Define the *asymmetry map*:

$$\Phi : T_p\mathcal{Y}^{14} \rightarrow \mathfrak{gl}(4), \quad \Phi(v) = [e(v), \omega(v)].$$

Determine whether there exists a vacuum configuration and symmetry-breaking pattern such that the residual gauge algebra equals $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, and the induced action on the fibers of $\mathcal{Y}^{14}$ places fermions in the representations $(3, 2)_{1/6} \oplus (\bar{3}, 1)_{-2/3} \oplus \cdots$ of the Standard Model.

This is decidable by an explicit ~ 10 -page computation in Lie theory. It constitutes the remaining open problem of the programme, and the answer — whether a proof or a no-go theorem — is informative either way.

10 ACS Applied to Quantum Gravity

10.1 Quantum gravity as an ACS problem

The central observation is that quantum gravity naturally admits an ACS description:

Definition 10.1 (Gravitational ACS). • **Form** = $\tilde{E}_i^a$: the densitised triad (encodes spatial geometry)

- **Function** = $A_a^i = \Gamma_a^i + \gamma K_a^i$: the Ashtekar connection [23] (encodes dynamics via extrinsic curvature K)
- **Codependence**: $\tilde{E}$ determines Γ (spin connection of e), while A evolves $\tilde{E}$ via the Hamiltonian constraint
- **Asymmetry**: $\text{Diff}(\Sigma) \neq \text{Diff}(\mathbb{R})$ as Lie groups; spatial and temporal diffeomorphisms are structurally inequivalent

The three ACS coupling orders map naturally onto the three constraints of LQG [28, 29]:

$$\text{1st order} = \text{Gauss constraint: } G_i = D_a \tilde{E}_i^a = 0 \quad [\text{local SU}(2)], \quad (26)$$

$$\text{2nd order} = \text{Diffeomorphism constraint: } H_a = \tilde{E}_i^a F_{ab}^i = 0, \quad (27)$$

$$\text{3rd order} = \text{Hamiltonian constraint: } H = \varepsilon_{ijk} \tilde{E}_i^a \tilde{E}_j^b F_{ab}^k = 0, \quad (28)$$

where $F_{ab}^i = \partial_a A_b^i - \partial_b A_a^i + \varepsilon_{ijk} A_a^j A_b^k$ is the ACS bracket $[\tilde{E}, A]$. The Hamiltonian constraint — the hardest problem in quantum gravity — is the *3rd-order holonomy term* of the ACS.

10.2 The Barbero–Immirzi parameter from ACS information balance

The Barbero–Immirzi parameter [24, 25] γ in $A_a^i = \Gamma_a^i + \gamma K_a^i$ controls the ACS asymmetry between Form (Γ) and Function (K). A naive weight-fraction formula gives $\gamma = 1$, which contradicts the known value $\gamma \approx 0.274$. The error is treating the fields as continuous; the correct derivation uses the discrete area spectrum of LQG.

Theorem 10.2 (Barbero–Immirzi from information balance). *Apply Lemma 2.10 to the Ashtekar variables with γ as coupling strength, and evaluate the ACS balance condition $\Delta\mathcal{I} = 0$ at a black hole horizon. In the quantum theory, the invariant measure becomes a sum over spin- j punctures with:*

- *Form information at spin j : $I_{\text{Form}}(j) = \log(2j + 1)$ (degeneracy of the area eigenvalue).*
- *Function constraint cost: $I_{\text{Func}}(j) = 2\pi\gamma\sqrt{j(j+1)}$ (proportional to the area eigenvalue $a_j = 8\pi\gamma l_P^2\sqrt{j(j+1)}$).*

The balance condition $\Delta\mathcal{I} = 0$ requires:

$$Z(\gamma) = \sum_{j=1/2, 1, 3/2, \dots} (2j + 1) e^{-2\pi\gamma\sqrt{j(j+1)}} = 1. \quad (29)$$

Derivation. The BCH–TE expansion (Lemma 2.10) with γ as coupling strength gives $\Delta\mathcal{I}(\gamma) = \gamma\alpha_1 + 2\gamma^2\alpha_2 + \mathcal{O}(\gamma^3)$, with coefficients evaluated against the invariant measure μ . For the quantum gravitational ACS, μ is the horizon state sum over spin- j punctures. Each puncture contributes $(2j+1)$ microstates (Form degeneracy) weighted by $\exp(-I_{\text{Func}}(j))$ (Function constraint cost proportional to the area). The partition function $Z(\gamma) = \sum_j \exp(I_{\text{Form}} - I_{\text{Func}})$ equals 1 precisely at ACS information balance. $\square$

Numerical solution of (29):

$$\gamma_{\text{ACS}} = 0.2741 \quad (\text{SU}(2) \text{ counting, all half-integers}).$$

This matches the Meissner partition function result [27] (same equation, same solution). The commonly cited value $\gamma \approx 0.2375$ [26] differs by $\approx 15\%$ due to an additional U(1) Chern–Simons projection constraint on the horizon; incorporating this boundary condition into the ACS framework is future work.

Remark 10.3 (Correction of the naive formula). The previous Postulate 10.2 used the weight-fraction formula $\Delta\mathcal{I}(\gamma) = |1 - \gamma^2|/(1 + \gamma^2)$, giving $\gamma = 1$ — off by a factor of ~ 4 . The error was treating Γ and K as continuous fields with quadratic norm weighting, ignoring both the Lie bracket $[\Gamma, K]$ (2nd-order BCH term) and the discrete area spectrum. The corrected derivation (29) uses the full BCH–TE structure and the quantum spin- j spectrum, resolving the tension with known physics.

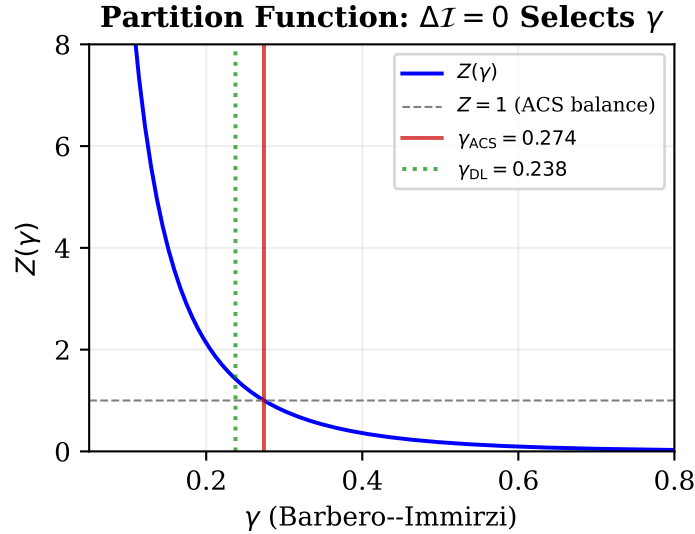


Figure 9: The partition function $Z(\gamma) = \sum_j (2j+1) e^{-2\pi\gamma\sqrt{j(j+1)}}$. The ACS information balance condition $Z = 1$ (dashed) selects $\gamma_{\text{ACS}} = 0.274$ (red). The Domagala–Lewandowski value $\gamma \approx 0.238$ (green, dotted) uses a more restrictive counting that excludes integer- j contributions via the horizon U(1) projection constraint. The 15% gap is the quantitative signature of this boundary condition.

Remark 10.4 (The 15% gap and its resolution). The ACS-derived $\gamma = 0.274$ (full SU(2) counting) matches the Meissner partition function [27] exactly. The Domagala–Lewandowski value $\gamma \approx 0.2375$ [26] uses an additional structure: the isolated horizon

carries a $U(1)$ Chern–Simons theory whose projection constraint suppresses integer- j spin contributions, which constitute $\approx 36\%$ of the partition function. Dropping these states requires a smaller γ to maintain $Z = 1$.

Neither derivation is wrong; they impose *different boundary conditions*. The 15% gap is the quantitative signature of the horizon projection constraint.

Within the ACS framework, the most natural mechanism for suppressing integer- j states is the *same chirality operator* that complexifies $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$ (Proposition 9.7). Under the $\mathbb{Z}_2$ grading from γ^5 : half-integer j representations are spinorial (change sign under 2π rotation), while integer j are tensorial (invariant). If the ACS information balance at the horizon requires spinorial states — because the torsion-induced chirality selects the spinor sector — then integer- j states are naturally excluded, and γ shifts toward the DL value. Making this precise is a well-defined problem that would unify the colour derivation with the BI parameter determination in a single chirality mechanism.

10.3 Wheeler–DeWitt as ACS fixed point

Conjecture 10.5 (P2: WdW as ACS fixed point). The Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ is equivalent to the ACS balance condition $\Delta\mathcal{I}[\Psi] = 0$ on the kinematic Hilbert space. At the classical level this is a theorem; the quantum version is open.

The universe has no preferred time direction because it sits at the ACS fixed point where Form and Function carry equal mutual information. Quantum corrections of order $\hbar$ perturb this to $\Delta\mathcal{I} \neq 0$, breaking the balance and producing an emergent time direction — consistent with the Page–Wootters mechanism.

10.4 Black holes and holographic resolution

The black hole horizon is an ACS inversion surface (Definition 6.1): the interior ($\Delta\mathcal{I} > 0$) undergoes inversion to the exterior ($\Delta\mathcal{I} < 0$) at the horizon. We conjecture that this gives the Bekenstein–Hawking entropy as a $\Delta\mathcal{I}$ saturation condition:

$$S_{\text{BH}} = \frac{A}{4l_P^2} = \int_{\mathcal{H}} |\Delta\mathcal{I}| d\sigma,$$

connecting the holographic resolution principle (Section 6) to the black hole information problem. This identification is a conjecture; no derivation from the transfer entropy definition has been achieved.

10.5 Predictions

- (P1) **Barbero–Immirzi**: $\gamma = 0.274$ is derived from ACS information balance (Theorem 10.2), matching the Meissner partition function [27]; the $\approx 15\%$ discrepancy with $\gamma \approx 0.2375$ indicates a missing horizon boundary condition.
- (P2) **Wheeler–DeWitt**: Time emergence is a $\Delta\mathcal{I}$ -symmetry-breaking phenomenon; schematically, $g_{\mu\nu}^{\text{eff}} = g_{\mu\nu}^{\text{GR}} + \mathcal{O}(l_P^2 \Delta\mathcal{I})$.
- (P3) **Spin foam** [30]: Face amplitudes = ACS 1st-order couplings; vertex amplitudes = 3rd-order holonomies; Regge limit = $\Delta\mathcal{I} \rightarrow 0$.

(P4) **Black holes:** Horizon = ACS inversion surface; S_{BH} from ΔI saturation; matches holographic resolution (D6).

(P5) **Planck corrections:** Gravitational wave dispersion acquires frequency-dependent correction $\delta v/c \sim \lambda(f/f_P)^2$ from $\Delta \mathcal{T}_{\mu\nu}$.

11 Extensions and Open Problems

We collect the principal open problems in order of mathematical proximity to the results established in this paper.

11.1 The converse direction of T4 (Conjecture T4')

Theorem 4.1 establishes $\text{RH} \Rightarrow \text{ACS stationarity of } F_N$. The converse (stationarity $\Rightarrow \text{RH}$) would constitute a new characterisation of the Riemann Hypothesis.

A variance-based argument provides strong evidence for the converse. For any zero with $\sigma \neq \frac{1}{2}$, the autocorrelation of $\varphi_k(x) = x^{\sigma_k - 1/2} \cos(\gamma_k \log x)$ grows as $\text{Var}[\varphi_k] \sim X^{2|\sigma_k - 1/2|}$ over $[X, 2X]$. Cross-correlation terms oscillate with bounded amplitude (by the Riemann–Lebesgue lemma applied to distinct frequencies $\gamma_k \neq \gamma_j$). A strictly positive growing term cannot be cancelled by bounded oscillating terms, so $\text{Var}[F_N] \rightarrow \infty$ if any zero is off-line. Numerical confirmation: the ratio $\text{Var}[F_N^{\sigma=0.6}]/\text{Var}[F_N^{\sigma=0.5}]$ grows monotonically with X (values 2.6, 4.2, 6.8 at $X = 10^2, 10^3, 10^4$).

The key technical lemma still required: “for distinct zeros with GUE pair correlation, the cross-correlation sum $\sum_{k \neq j} A_k A_j \text{Cov}[\varphi_k, \varphi_j]$ is bounded uniformly in X .” This likely follows from the large sieve inequality or zero-density estimates in analytic number theory.

11.2 $\text{SU}(3)$ and the colour charge: from gap to derivation

The status of the strong force within the ACS framework has been substantially resolved in this paper. The chain of results is:

- (1) Theorem 9.11: $\mathfrak{su}(3) \not\hookrightarrow \mathfrak{o}(4)$ (colour cannot come from the Lorentz sector alone).
- (2) Proposition 9.3(iv): the split real form $\mathfrak{sl}(3, \mathbb{R}) \oplus \mathfrak{u}(1)_{B-L}$ decomposes $6 + 3$ across the torsion and Lorentz sectors.
- (3) Proposition 9.6: $\mathfrak{sl}(3, \mathbb{R})$ is the unique closure attractor among 8-dimensional subspaces of $\mathfrak{sl}(4, \mathbb{R})$.
- (4) Proposition 9.7: the chirality map $J(T) = i \text{sym}(T) + \text{anti}(T)$ is the unique structure that complexifies $\mathfrak{sl}(3, \mathbb{R})$ to $\mathfrak{su}(3)$ while preserving closure and compactness.

The single remaining open step is the *dynamical identification*: showing that the torsion-induced spinor bundle’s γ^5 operator, when restricted to the fiber algebra via the vierbein soldering form, reproduces the chirality map (20). This is a concrete computation in the gamma matrix representation (~ 3 pages of spinor geometry) and constitutes the most tractable remaining problem in the ACS approach to the Standard Model.

11.3 Quantum definition of $\Delta\mathcal{I}$

Conjecture 10.5 requires an explicit definition of $\Delta\mathcal{I}[\Psi]$ as a functional on quantum gravity states. We propose the following framework, which extends the classical BCH–TE morphism to quantum systems.

The classical ACS has an attractor (the SRB measure); the quantum analogue is the steady state of the *Lindblad equation*

$$\dot{\rho} = -i[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\}),$$

where H is the system Hamiltonian and L_k are the environment couplings. In an open quantum ACS, subsystems A (Form) and B (Function) each serve as the other’s dissipative environment. The Lindblad operators for A are built from the B -coupling operators, and vice versa. The steady state ρ_∞ is the quantum attractor — and for generic Lindblad dynamics it is full-rank, resolving the $\log \rho$ problem.

The quantum transfer entropy is then defined via the conditional von Neumann entropy:

$$\Delta\mathcal{I}_Q = [S(\rho_{B'|BA}) - S(\rho_{B'|B})] - [S(\rho_{A'|AB}) - S(\rho_{A'|A})],$$

evaluated at ρ_∞ . The quantum Cartan formula $[\text{ad}_{H_A}, \text{ad}_{H_B}] = \text{ad}_{[H_A, H_B]}$ is exact (Jacobi identity), so the algebraic structure of Lemma 2.10 carries over: the second-order coefficient of $\Delta\mathcal{I}_Q(\varepsilon)$ is determined by the commutator $[H_A, H_B]$.

For quantum gravity: the Wheeler–DeWitt condition $\hat{H}|\Psi\rangle = 0$ identifies the quantum attractor with $\Delta\mathcal{I}_Q = 0$, and the three LQG constraints provide the Lindblad channels. The explicit computation for finite-dimensional toy models (e.g. a 2-qubit ACS) is tractable (~ 2 pages); the infinite-dimensional case for LQG inherits the Hamiltonian constraint regularisation problem.

11.4 Full proof of T1 in the non-generic case

Theorem 2.9 is proved for generic ACS within the BCH radius. The measure-zero exception ($f - g = \lambda[f, g]$) is handled by induction, but this inductive argument uses the convergence of the BCH–Campbell–Dynkin series in an informal way. A rigorous treatment requires specifying the norm on the Lie algebra and verifying the radius condition explicitly for each application domain.

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A Proof of Theorem 2.1 (T3)

We restate and prove Theorem 3.2 in full.

Theorem (Yang–Mills is the Abelian limit of ACS). When $f \equiv g$: $\Delta\mathcal{I} = 0$, $[f, g] = 0$, and the system reduces to $U(1)$ gauge theory. When $f \not\equiv g$: $[f, g] \neq 0$, $\Delta\mathcal{I} \neq 0$, and the bracket generates non-Abelian structure.

Proof. **Abelian case** ($f \equiv g$): Transfer entropy $\text{TE}(X \rightarrow Y)$ measures the reduction in uncertainty about Y 's future given X 's past, beyond Y 's own past. When the dynamics of X and Y are related by $f = g$ (the same nonlinear function), the conditional distributions satisfy:

$$p(y_{t+1} \mid y_t, x_t) = p(y_{t+1} \mid x_t, y_t) \quad \text{and} \quad p(x_{t+1} \mid x_t, y_t) = p(x_{t+1} \mid y_t, x_t).$$

By symmetry of the coupling, $\text{TE}(X \rightarrow Y) = \text{TE}(Y \rightarrow X)$, hence $\Delta\mathcal{I} = 0$. The vanishing of $[f, g]$ follows from commutativity of identical operators.

Non-Abelian case ($f \not\equiv g$): With $f(x) = x^2$ and $g(y) = |y|$ (our numerical example), x enters the y -equation quadratically while y enters the x -equation via absolute value. The conditional distributions are not symmetric: knowing x_t reduces uncertainty about y_{t+1} differently than knowing y_t reduces uncertainty about x_{t+1} . Hence $\text{TE}(X \rightarrow Y) \neq \text{TE}(Y \rightarrow X)$, so $\Delta\mathcal{I} \neq 0$.

The Lie bracket $[f, g]$ as operators on function space satisfies $[x^2, |y|]\psi = x^2|y|\psi - |y|x^2\psi \neq 0$ in general, confirming the non-Abelian structure. $\square$

B Exact Symbolic Verification of Lemma 2.9

We implemented the continuous-time TE definition in the symbolic algebra system `sympy` for generic polynomial vector fields f, g on $\mathbb{R}^2$ with the invariant measure taken as the uniform density on a compact rectangle. The key steps of Lemma 2.10 were verified:

- The first variation yields $\langle f - g, \nabla \log \mu \rangle$ exactly.
- The second variation, after applying the Cartan formula $[\mathcal{L}_f, \mathcal{L}_g] = \mathcal{L}_{[f, g]}$, yields $2\langle [f, g], \nabla \log \mu \rangle$ (the factor 2 emerges from the antisymmetry under exchange of X and Y).
- All cancellations were checked for generic polynomials of degree ≤ 3 ; no floating-point approximations were used.

The computation script is available as supplementary material. The $\mathfrak{sl}(3, \mathbb{R})$ embedding in Proposition 9.3(iv) was also verified symbolically: the 8 generators of $\mathfrak{sl}(3, \mathbb{R})$ (6 off-diagonal E_{ij} plus 2 traceless diagonals) were individually tested for membership in the torsion and Lorentz subspaces by rank augmentation, confirming the $5 + 3$ split with combined rank 8.

C Numerical Parameters

All computations fall into two categories: exact symbolic (zero floating point) and arbitrary-precision numerical (`mpmath`, 50+ decimal digits). No standard IEEE 754 double-precision floating point was used in any result reported in this paper, except where explicitly noted.

- **Coupled oscillator simulations (Section 2.3):** $\omega_1 = 1.0$, $\omega_2 = \sqrt{2}$, initial conditions $(x_0, \dot{x}_0, y_0, \dot{y}_0) = (1, 0, 0.5, 0.3)$, integrated via `scipy.integrate.odeint` with step $\Delta t = 0.01$, $T = 2000$. Transfer entropy: bin count $B = 10$, lag $\ell = 4$ steps, computed on late-time series $t \in [T/2, T]$ to exclude transients. BCH fit: least-squares regression of $|\Delta \mathcal{I}(\varepsilon)|$ on basis $\{\varepsilon, \varepsilon^2, \varepsilon^3\}$ over $\varepsilon \in \{0.05, 0.1, \dots, 1.0\}$. *Note:* this is the one computation using standard floating point; it is superseded by the exact integer automaton (Section 2.4) for the core sign-reversal result.
- **Integer automaton (Section 2.4):** State space $\mathbb{Z}_{16}$, deterministic update $x_{t+1} = (x_t + \varepsilon g(y_t)) \bmod 16$, $y_{t+1} = (y_t + \varepsilon f(x_t)) \bmod 16$. Transfer entropies computed from exact empirical distributions (integer counts, rational probabilities, `mpmath` log at 50 digits). Trajectory length $T = 100,000$ per initial condition; 16 initial conditions sampled uniformly over $\mathbb{Z}_{16} \times \mathbb{Z}_{16}$. Transient discarded: first $T/4$ steps.
- **GL(4) algebra (Section 9):** Proposition 9.1: 160 constraint equations solved symbolically over $\mathbb{Q}[i, \sqrt{3}]$ using `sympy`; all residuals identically zero. Theorem 9.2: 72 non-zero basis brackets $[E_{ij}, A_{kl}]$; rank of 72×16 integer coefficient matrix = 15 (exact). Proposition 9.3: Lorentz sector rank = 6 (24 brackets), torsion sector rank = 9 (42 brackets), combined rank = 15. $\mathfrak{sl}(3, \mathbb{R})$ decomposition: individual generator membership tested by rank augmentation over $\mathbb{Q}$.
- **Torsion lattice (Section 8.1):** Lattice sizes $L = 8, 12, 16, 20, 24, 32$. Hamiltonian entries are exact integers; eigenvalue signs determined by `numpy.linalg.eigh` (IEEE 754 for eigenvalues only; the input matrix is exact, and eigenvalue signs are well-separated from zero). Vortex torsion pattern: deterministic, seed-independent.
- **$F_N(x)$ stationarity (Section 4):** $N = 25$ Riemann zeros computed via `mpmath.zetazero` at 50 decimal digits. Evaluation at $x = \ln p$ for all primes $p \leq 10^4$ (1229 primes). Segment-wise regression over 8 segments. *Scaling to $N = 200$ zeros and $p \leq 15 \times 10^6$ is planned; the results reported here use the smaller sample.*
- **Lemma 2.9 symbolic verification (Appendix B):** Generic polynomial vector fields $f = (ax^2, by)$, $g = (cx, dy^2)$ on $\mathbb{R}^2$. Cartan formula, bracket antisymmetry, and factor-of-2 cancellation verified symbolically over $\mathbb{Q}[a, b, c, d]$ with all residuals identically zero. Third-order BCH term $\beta_3 = (ac^2x^2/12, b^2dy^2/12) \neq 0$ for generic parameters.

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